

‘Off and Away All Day’ Smartphone Use and Mental Health

Evidence from the staggered rollout of mobile phone bans in
Australian schools



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A thesis presented for the degree of
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Abstract

Recent years have seen mounting evidence of a substantial deterioration in adolescent mental health across the Western world. Among the potential drivers of this decline, the rapid proliferation of smartphones and social media has attracted increasing scrutiny. This research examines a prominent policy response to such concerns: mobile phone bans in schools. It employs a difference-in-differences research design, leveraging the staggered rollout of bans across Australian states beginning in 2020. This natural experiment enables causal estimation of the impact of an abrupt halt in smartphone use during the school day. Event study estimates suggest that phone bans yielded modest improvements in student mental health, though these effects are small in magnitude and emerge only in later years. Additional evidence on individual mental health dimensions largely points to negligible effects. However, male students show improvements in their capacity to engage in social activity.

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Keywords: Mental Health, Smartphone Bans, Difference-in-Differences

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Chapter 1

Introduction

Since the advent of the internet in the 1990s, the world has witnessed a remarkable shift towards mobile connectivity. The spread of internet-connected devices has facilitated the consumption of digitised content, reshaping behavioural patterns and sometimes crowding out offline pursuits. Smartphone adoption, in particular, has grown sharply over the last fifteen years, with the number of mobile phone subscriptions recently surpassing the global population ([World Economic Forum, 2023](#)). As social interactions and entertainment have migrated online, screen time has surged. This is especially apparent among those born after 1995, the first generation to navigate adolescence with pervasive access to internet-connected devices. By 2017, 94% of Australian teenagers owned a smartphone, with the average user spending more than four and a half hours per day on their device ([Rhodes, 2017](#)). In the United States, 13- to 19-year-olds averaged nearly five hours per day on social media alone ([Gallup, 2023](#)).

As smartphone use has increased, reported rates and diagnoses of psychiatric disorders among children and adolescents have risen across Western countries ([Collishaw, 2015](#)). In the United States, for example, the number of adolescents exhibiting high levels of depressive symptoms increased by 33% between 2010 and 2015 ([Twenge et al., 2018](#), p. 8). Similarly, in Australia, the proportion of 15- to 19-year-olds experiencing high or very high psychological distress increased by ten percentage points in the decade leading up to 2022 ([McHale et al., 2023](#)).

Adolescence is commonly understood as the transition period from the onset of puberty to adulthood and parental independence. While it is traditionally viewed as ending around age 18, recent research suggests it may start as early as age 10 and extend to age 24. This is in line with contemporary views on pubertal timing and social role transitions ([Sawyer et al., 2018](#)). As a critical developmental stage, adolescence marks a period of heightened vulnerability to psychological distress, with a majority of

lifetime mental disorders having their onset before it comes to a conclusion (McGorry et al., 2024). The effects of such early disorders often extend into adulthood, shaping health trajectories across the lifespan (Patton et al., 2016).

Given the detrimental repercussions of mental illness on many areas of young people's lives, including their education and physical health, its prevention is a critical public policy priority. Addressing this challenge requires a clear understanding of the drivers behind the recent deterioration in youth mental health.

Scholars and policymakers have hypothesised that the rise of mobile connectivity and social media use in particular might be a contributing factor (Haidt, 2024; Parliament of the Commonwealth of Australia, 2024). Governments have quickly responded with new regulations, aiming to shield adolescents from the excessive use of digital devices. Mobile phone bans in schools are a prominent example. Recent years have seen a wave of such policies emerge, including a nationwide phone ban in French middle schools in 2018, and subsequent measures in parts of Spain, Canada, Australia, and the United States. Based on the view that smartphones contribute to distraction, cyberbullying, and lower mental health, school phone bans are taken to produce the reverse outcomes: improved learning, reduced bullying, and better adolescent well-being. Well-identified causal evidence underpinning these expectations, however, remains scarce.

In this thesis, I provide quasi-experimental estimates of the impact of smartphone use on school student's mental health by exploiting a natural experiment: the staggered rollout of school mobile phone bans across Australian states starting in 2020. Coupling longitudinal data on the mental health of Australian school students with a difference-in-differences strategy, I find limited evidence that mobile phone bans improved student mental health. While standard two-way fixed effects regressions produce null results, event study estimates suggest dynamically increasing effects, which turn significant in the third year after the implementation of bans. I also find that bans reduced the extent to which male students reported emotional or physical health interfering with social activities.

The expansion of school phone bans across Australian states offers a promising setting to examine the impact of smartphone use on adolescent mental health. Three of eight states introduced bans in early 2020, followed by three more in 2023. In 2024, as the remaining two states followed suit, the policy effectively became nationwide. The bans are strict: branded under the slogan "Off and away all day", they prohibit mobile phone use across year levels, including during breaks, and extend to smartwatches and other communication devices.

My analysis uses data from the *Household, Income and Labour Dynamics in Australia* (HILDA) survey, a nationally representative longitudinal study of around 17,000 households, launched in 2001. The latest release, published in January 2025, contains 23

waves, with observations up to February 2024, shortly after phone bans were extended to all public schools across Australia. Personal interviews are conducted with household members aged 15 and above. Critically, these interviews include a clinically validated mental health questionnaire: the SF-36 Health Module.

My difference-in-difference design exploits two sources of variation: a student's state of residence, and whether they were interviewed before or after mobile phone bans took effect in their school. Under no anticipation and parallel trends assumptions, this variation enables causal identification of mental health effects.

A challenge to my identification strategy arises from the COVID-19 pandemic: it overlaps with my sample period and had well-documented adverse effects on adolescent mental health (Li et al., 2022). Its intensity varied across states, and Australia's federal structure led to heterogeneous policy responses. I address this threat to the parallel trends assumption in three ways. First, I draw on data from HILDA's COVID modules, which include questions on the extent to which a respondent's life was disrupted by the pandemic. Second, I incorporate biweekly COVID mortality data by state as a proxy for the severity of local public health responses. Third, I compile dates of COVID-related school closures by state and year level from government press releases. This allows me to account for differences in the duration of students' exposure to mobile phone bans.

Separately, to address econometric concerns about bias in two-way fixed effects estimations of staggered adoption designs, I estimate an event study specification using an alternative robust estimator.

This thesis remains largely silent on the mechanisms through which smartphone use may affect mental health. Existing research in economics and psychology points to several plausible channels. These include improved academic performance, reduced phone addiction, a reallocation of time towards mental health-enhancing activities such as face-to-face interactions, and a decline in the tendency to engage in unfavourable social comparisons, which may be amplified by social media consumption in particular (Beland and Murphy, 2016; Allcott et al., 2020; Braghieri et al., 2022).

The analysis presented in this thesis should be interpreted with caution. First, the findings pertain to a narrow sub-population: namely, 15- to 18-year-old students enrolled in public schools. They may therefore overlook potentially stronger effects among younger children, who are likely more vulnerable to the mental health risks of smartphones (Orben et al., 2022). Second, my analysis does not take into account how phone bans reshaped screen time outside of school hours. For example, if students compensate for in-school restrictions by increasing their phone use when at home, then the estimated effects cannot be straightforwardly interpreted as the consequence of a simple reduction in exposure, but rather as reflecting a shift in the temporal and

contextual pattern of digital engagement. Third, the analysis does not account for school-level mobile phone policies that preceded statewide bans.

This thesis contributes to a growing literature in economics and psychology which aims to identify the causal effect of internet and social media on mental health. It adds to this literature in several ways. First, it employs a more granular and detailed measure of mental health than is common in existing studies. Second, instead of examining adult populations, as much of the literature has done, my sample consists of adolescents, a demographic that has experienced the steepest decline in mental health over the past two decades. Third, this thesis examines a setting in which smartphone exposure comes to an abrupt halt due to a policy intervention. This stands in contrast to existing studies, which primarily investigate the mental health effects of rising digital device use over time. The directionality of the shock matters: there is no guarantee that the effects of reducing smartphone use will be symmetric to those of increasing it. Long-term exposure over the last decade may have reshaped adolescents' time use, social habits and even cognitive processes in ways that are not easily reversed. If digital engagement has displaced activities with positive mental health returns, such as in-person interaction or community involvement, then reducing exposure alone may not automatically restore these activities, or their associated benefits.

The remainder of the thesis is organised as follows: Chapter 2 reviews the existing literature, with a focus on recent work that aims to establish causal effects; Chapter 3 provides background on mental health trends and the rollout of mobile phone bans in Australia, describes the data sources, and presents descriptive statistics; Chapter 4 discusses the empirical strategy; Chapter 5 presents the results; Chapter 6 concludes.

Chapter 2

Literature Review

Studying the link between smartphone use and mental health is challenging for several reasons. First, the direction of the relationship is unclear: while smartphones may influence mental health outcomes, it is equally plausible that changes in mental health drive subsequent changes in smartphone use. Second, it is challenging to study the effects of an exposure on an outcome if the exposure is nearly omnipresent. For instance, with over 90% of Australian adolescents owning a smartphone, identifying unexposed individuals, or comparison groups with meaningfully lower exposure, is difficult. Third, smartphone use is not monolithic. On the contrary, smartphones enable a wide range of activities, each of which may have distinct mental health implications. For example, messaging distant relatives may be beneficial, while late-night social media scrolling may be harmful. Neither activity would be possible without smartphones, yet mere screen time metrics do not discriminate between them. Screen time is therefore a somewhat coarse and behaviourally uninformative independent variable. Fourth, the literature proposes a range of mechanisms linking smartphone use and mental health, without a dominant one emerging. Some of these mechanisms may work against one another. For example, online interactions can foster inclusion for marginalized groups, while simultaneously displacing face-to-face interactions with family. Fifth, empirical work points to considerable heterogeneity of health effects across demographic groups, limiting the generalisability of findings beyond narrow subpopulations.

Despite these challenges, the question of the relationship between smartphone use and mental health has received treatment in both the psychology and economics literature. Much of this research focuses on the effects of digital media consumption in general, where digital media refers to any digitised content transmitted via the internet. Within digital media, social media has received particular attention. Few studies distinguish between the various internet-connected devices through which such content is accessed. To date, no clear consensus on the mental health effects of digital

media or its components has emerged. The remainder of this section reviews findings from both psychology and economics.

2.1 Overview of the Psychology Literature

The psychology literature generally reports small and statistically insignificant effects, both positive and negative. This body of work is limited by several methodological problems, including an over-reliance on cross-sectional designs and a lack of robust identification strategies for causal inference.

Several studies in the field have hypothesised that adolescents' constant digital connectivity is driving recent increases in depression, anxiety and suicide rates, particularly among girls, who are the heaviest users of digital media (Mojtabai et al., 2016; Twenge et al., 2018). This hypothesis has since been questioned on the grounds that observed correlations between media use and depressive symptoms are very small, prompting alternative explanations such as the prolonged effects of the Great Recession or rising academic pressure (Daly, 2018).

In an early review, Best et al. (2014) analyse 43 studies published between 2003 and 2012 which focus on the association between digital media use and mental health in adolescent or child populations. They report contradictory evidence of negative, positive or null results, and emphasise the lack of robust causal research designs. The authors also identify a range of hypothesised mechanisms: some beneficial, such as opportunities for social connection and identity experimentation; others detrimental, including increased social comparison and the displacement of face-to-face interaction.

A more recent review synthesises 13 studies focused on adolescents aged 13 to 18, each using granular metrics of social media use and clinically validated measures of psychological distress (Keles et al., 2020). The authors report an association between increased use of digital technologies and mental health symptoms, which they emphasise should be interpreted as correlational rather than causal. Twelve of the thirteen studies are cross-sectional, and inconsistencies across results likely reflect the omission of relevant moderating variables. In sum, the observed association is likely contingent on third factors, and sensitive to variations in usage patterns, pre-existing vulnerabilities and the context of technology engagement.

A common limitation across these studies is their reliance on self-reported screen time, which is prone to recall bias. Meta-analytic evidence suggests that the correlation between actual and self-reported screen time is weak, estimated at around 30% (Parry et al., 2021). When combined with self-reported mental health outcomes, which are

susceptible to common method bias, this casts doubt on the robustness of the resulting conclusions (Podsakoff et al., 2003).

Consistent with the meta-analyses described above, recent large-scale and preregistered studies in psychology have not yielded robust evidence linking digital media use to adolescent mental health. Notably, Orben and Przybylski (2019) apply specification curve analysis across three national data sources of adolescents from the U.S. and UK. They show that different specification choices, including how technology use and well-being are measured, as well as the selection of covariates, generate a wide range of estimated effect sizes. The median standardised effect across more than 3 billion model specifications is negative but small, with technology use accounting for 0.4% of the variation in well-being. As with prior studies, this cross-sectional association provides no credible basis for causal inference, nor for disentangling directionality.

A recent follow-up study by the same authors and colleagues finds that the cross-sectional relationship between digital media use and psychological well-being varies across the life span, and undergoes key changes during adolescence (Orben et al., 2022). Biological, psychological and social changes during this period may heighten sensitivity to both the positive and negative effects of digital technology and social media. Using longitudinal data from the UK, the authors identify distinct developmental windows of sensitivity: between ages 11 and 13 for girls, and 14 to 15 for boys.

Finally, diary and ecological momentary assessment studies, which track adolescents intensively to generate high-frequency data on technology use and well-being, also yield inconclusive results. These designs reduce recall bias and allow researchers to capture individual usage patterns more accurately. For example, Jensen et al. (2019) analyse data from a sample of 2,100 adolescents who reported psychological well-being three times per day, and their technology use once per day. The authors find no evidence of an association between reported technology use and mental health.

In sum, work in psychology on digital media use and adolescent mental health has been largely correlational. These studies do not adequately address concerns about reverse causality and fail to account for confounding factors. Notwithstanding, several valuable insights emerge. First, findings are highly sensitive to how the dependent and independent variables are measured. For instance, though screen time captures the intensity of technology use, it is prone to recall bias and fails to reflect the diversity of online experiences. Moreover, results may be severely distorted by functional form misspecification. Second, effects, assuming they exist, may vary across an individual's life span, and should not be extrapolated beyond the subgroups on which they are estimated.

2.2 Overview of the Economics Literature

In economics, research has primarily focused on the mental health effects of social media use and internet access, with relatively few studies taking mobile phones or mobile connectivity as their primary object of interest. However, disentangling the separate effects of these technologies is generally not feasible, especially in observational data, owing to their interdependence: mobile phones enable internet access, which in turn enables social media use. Given that American adolescents aged 13 to 19 average 4.8 hours of social media use per day, most of it via smartphones, any estimate of the mental health effects of social media likely reflects those of smartphone use more broadly, and vice versa (Gallup, 2023).

2.2.1 Experimental Studies

A first set of economics studies offering credible causal estimates of social media's impact on psychological well-being relies on experiments that incentivize a randomly selected subset of participants to reduce their social media use.

Allcott et al. (2020) recruit 2,743 Facebook users via online ads, eliciting their willingness-to-pay for deactivating their accounts for one month in the lead-up to the 2018 U.S. mid-term election. Participants were then randomly assigned to a treatment group, which was paid to deactivate, and a control group, which was not. Compliance was monitored through regular checks of treated individuals' public profile pages. Using text messages, surveys and emails, the authors measure a variety of outcomes, including self-reported happiness, life satisfaction, depression and anxiety, as well as time-allocation. Deactivation led to a small but significant improvement in subjective well-being, with an effect size equivalent to 25-40 percent of that typically observed in psychological interventions such as self-help therapy or group training. However, effects on subjective well-being, as captured by responses to daily text messages, were insignificant. Finally, treated individuals did not substitute Facebook use with other online activities. Instead, they reallocated time towards watching television and face-to-face interactions.

Similarly, Mosquera et al. (2020) enrol 1,769 individuals from a U.S. university in a field experiment, randomly assigning a subset to a treatment group that is paid to deactivate Facebook for one week. Perhaps unsurprisingly, given the short treatment duration, the authors find no significant effects on life satisfaction. However, they do observe a 0.57 point reduction (on a 0-10 Likert scale) in self-reported symptoms of depression, with effects mostly driven by men.

Allcott et al. (2022) investigate digital addiction and the extent of self-control problems in digital media use. They recruit 2,000 American adults and ask them to install

an application which tracks screen time and allows users to set screen time limits. A randomly selected subset of participants is assigned to a treatment group which receives a \$2.50 subsidy for each hour of reduced social media use over a three-week period between two surveys. Treated individuals report higher subjective well-being, the effect being statistically significant but small. Changes in measures of happiness, life satisfaction, anxiety and depression, however, are statistically insignificant.

Finally, in what is the largest experimental study to date on the emotional effects of social media deactivation, [Allcott et al. \(2025\)](#) recruit 19,857 Facebook users and 15,585 Instagram users, randomly assigning 27 percent to a treatment group paid to deactivate their accounts for six weeks prior to the 2020 U.S. election. Deactivation causes a 0.06 standard deviation improvement in an index of happiness, anxiety, and depression among Facebook users. Among Instagram users, the effect lies at 0.041 standard deviations. The Facebook effect is largely driven by users over 35; the Instagram effect is driven by women under 25. Contrary to the evidence cited above, treated users almost fully substitute the time freed by Facebook or Instagram deactivation with other smartphone apps.

In sum, experimental studies suggest that social media deactivation causes small but statistically significant improvements in subjective well-being. However, several limitations constrain the generalisability of these findings. First, measures of well-being vary across studies, complicating direct comparisons. Second, samples may be unrepresentative due to selection mechanisms. Third, experimenter demand and Hawthorne effects may influence estimates. Fourth, all studies described above study short-term disruptions, lasting as little as one week in one case, which cannot address longer-term or dynamic effects. Fifth, most experimental work targets adult populations, and therefore offers little insight into the role of smartphones and social media in the deterioration of adolescent mental health.

2.2.2 Natural Experiments

A second set of economics papers exploits natural experiments, allowing researchers to adopt a general equilibrium perspective and to study longer-term or dynamic effects, thereby addressing some of the limitations inherent in experimental designs.

Using data from the *Understanding Society* panel, which covers over 40,000 UK households, [McDool et al. \(2020\)](#) estimate a fixed effects model with average neighbourhood broadband speed as the main independent variable, arguing that it serves as proxy for household internet use. Under the assumption that broadband speed is distributed quasi-randomly across space, they are able to estimate a causal parameter: the intention-to-treat effect of broadband speed on psychological well-being. To support their identification assumption, the authors note that broadband rollout was financed

through a government-subsidised programme, and does not systematically relate to neighbourhood socio-economic characteristics. They find that a one percent increase in local average broadband speed reduces self-reported satisfaction with appearance among 13- to 15-year-old teenagers by 0.6 percent, with a smaller effect on overall life satisfaction. Effects are notably stronger among girls than boys.

[Golin \(2022\)](#) relies on data from the German Socio-Economic Panel, focusing on respondents in the former Federal Republic of Germany. To capture internet access, Golin uses 2008, 2010 and 2012 survey modules on broadband adoption. Her outcome variable is based on the SF-12 health scale, a shortened version of the questionnaire that will also be used in this thesis. Since broadband adoption may be correlated with unobserved factors which also affect mental health, Golin employs household distance from the nearest telecommunications distribution frame as an instrumental variable. She argues that these frames, built in the 1960s and placed largely based on logistical convenience, are quasi-randomly distributed across space. Moreover, household distance to the nearest distribution frame is strongly correlated with internet access, satisfying the relevance condition for a valid instrument. Golin finds that the arrival of broadband in Germany between 2008 and 2012 negatively affected the mental health of women aged 17 to 30, reducing SF-12 scores by 0.149 standard deviations, but had no statistically significant effects for other groups.

Proceeding along similar lines, [Donati et al. \(2022\)](#) combine municipal-level data sources on broadband and 3G availability in Italy from 2001 to 2013. They merge this with patient-level data on mental health diagnoses taken from hospital discharge reports. As an instrumental variable, they use the distance from the centroid of each municipality to the nearest higher-order telecommunications exchange. As in the case of [Golin](#), the installation of these exchanges predates the advent of broadband and is argued to be plausibly exogenous. The authors find significant and sizeable effects among cohorts born between 1985 and 1995: moving from 0 to 100% broadband coverage in a province is estimated to increase diagnoses of mental health conditions by 111%. The effect is similar across genders but is concentrated among individuals exposed before the age of 20.

Finally, and most closely related to this thesis, [Braghieri et al. \(2022\)](#) exploit the staggered rollout of Facebook to U.S. colleges to estimate the causal effect of social media on mental health using a difference-in-differences strategy. From February 2004 to September 2006, Facebook was gradually introduced to 775 colleges before being made available to the general public. The authors show that after being given access, colleges experienced broad and rapid Facebook adoption among students. They find that the introduction of Facebook had negative consequences on self-reported mental health. Benchmarking the magnitude of effects, they suggest it is equivalent to approximately 22 percent of the effect of a job loss. Moreover, the authors provide

evidence that effects are driven by social media encouraging unfavourable social comparisons. In support of this hypothesis, the authors show that students who may be more prone to suffer from social comparisons – e.g., those living off campus – display worse effects. Furthermore, they address a commonly cited alternative explanation: that increases in reported psychiatric problems reflect reduced stigma, rather than actual declines in mental health. Arguably, Facebook lowered stigma surrounding mental health issues, thereby increasing individuals' willingness to report symptoms. Against this view, the authors show that Facebook's introduction did not reduce the share of missing answers to survey mental health questions, nor did it increase reporting of other stigmatised conditions, such as sexual assault. If reduced stigma were a major confounding factor, both of these patterns would plausibly have played out.

To summarise, the economics literature largely converges on the view that digital media consumption has either negative or null effects on mental health, depending on the subpopulation studied. Consistent with correlational findings in psychology, there is evidence that younger individuals and women may be particularly vulnerable. However, the relative importance of underlying causal mechanisms remains poorly understood. Moreover, due to data limitations, existing work has yet to adequately capture the diversity of individuals' online experiences.

Chapter 3

Background and Data

3.1 Background

3.1.1 Adolescent Mental Health in Western Societies

The American Psychological Association defines mental health as "a state of mind characterized by emotional well-being, good behavioural adjustment, relative freedom from anxiety and disabling symptoms, and a capacity to establish constructive relationships and cope with the ordinary demands and stresses of life" ([American Psychological Association, 2020](#)). This definition underscores the multi-dimensionality of the concept, and foreshadows the substantial difficulty involved in measuring it. Commonly used indicators, such as suicide rates, hospitalisations, or self-reported use of mental health services, capture only fragments of this broader construct, and therefore imperfectly track changes in population-level mental health. For instance, suicide rates reflect the incidence of acute and extreme distress without capturing an individual's broader "freedom from anxiety and disabling symptoms." A comprehensive picture thus requires a range of complementary indicators.

Over the past two decades, reported rates and diagnoses of psychiatric disorders have risen markedly across many Western countries, including Australia, the U.S., Greece, Germany, Sweden, Iceland, Norway, China, and New Zealand ([Collishaw, 2015](#)). These observed trends may be shaped in part by a shift in how symptoms are perceived and reported. Help-seeking by parents and children has become more common; screening and clinical recognition in schools have improved; diagnostic classifications have broadened, and some emotional states and behaviours previously considered normal have been medicalised ([Dowrick and Frances, 2013](#)). Nonetheless, there is mounting evidence of real increases in population symptom prevalence. Cross-cohort comparisons, for instance, which use equivalent mental health screening methods across unselected population samples, point to increases in adolescent depression

and anxiety (see, e.g., [Collishaw et al., 2010](#)). A recent report by the *Lancet Psychiatry Commission* finds substantial evidence of worsening youth mental health since the early 2010s, based on data from Australia, the U.S., the UK, and Denmark. It documents sharp increases in anxiety, depression, psychological self-harm and suicide, and further notes that this deterioration has accelerated with the COVID-19 pandemic ([McGorry et al., 2024](#)).

In Australia, the prevalence of operationally defined mental disorders among 16- to 24-year-olds rose by 50% between 2007 and 2021 ([Australian Bureau of Statistics, 2022](#)). The *Young Minds Matter survey*, conducted in 1998 and again in 2013-2014, paints a concerning picture of the mental health of Australian school-aged children ([Lawrence et al., 2015](#)). Based on responses from 76,606 households, the 2014 wave found that 13.9% of 4- to 17-year-olds had a mental disorder in the previous 12 months, and 17% had used services for emotional or behavioural issues. Among adolescents aged 11 to 17, 19.9% displayed high or very high levels of psychological distress, and 7.7% met the diagnostic criteria for major depressive disorder in the past year. Additionally, 10.9% of 12- to 17-year-olds reported having ever self-harmed, and 2.4% reported a suicide attempt in the previous 12 months. Compared to the 1998 survey, the prevalence of major depressive disorder rose by 52%.

Australia is no exception: in the United Kingdom, hospital self-harm presentations of 13- to 16-year-old girls increased by 68% between 2011 and 2014 ([Morgan et al., 2017](#)). Similarly, in the U.S., suicide rates have increased across all age groups, with the sharpest rise among girls aged 10 to 14, where the rate tripled between 1999 and 2017 ([Hedegaard et al., 2018](#)). Globally, the burden of mental illness continues to grow ([Murray et al., 2012](#)): in 2021, approximately 14% of individuals aged 10 to 19 suffered from mental health problems, comprising 13% of all health conditions in this age group ([World Health Organization, 2021](#)). Suicide was the third-leading cause of death among 15- to 29-year-olds ([World Health Organization, 2025](#)). Notably, experiences of mental health disorders in early life are linked to poorer mental health in adulthood ([Copeland et al., 2014](#)).

Adolescence is a sensitive period for the emergence of mental illness, with around 75% of all cases beginning before the age of 25 ([Solmi et al., 2022](#)). Spanning the ages of 10 to 24 ([Sawyer et al., 2018](#)), it marks a time of major developmental change which increases vulnerability to psychological distress. Physical and chemical changes in the brain heighten emotional responsiveness to both real and perceived stressors ([Bailen et al., 2019](#)) at a time when the regulatory system necessary to manage these emotions is not fully developed ([Somerville et al., 2010](#)). Adolescence is also marked by increased social sensitivity ([Somerville, 2013](#)). Friends begin to replace parents as the primary social contacts, making peer relationships a central source of conflict and stress. Acceptance and rejection by peers become critical to adolescents' sense of self-

worth (Connell and Wellborn, 1991), with negative peer interactions often leading to symptoms of anxiety and depression (La Greca and Harrison, 2005).

When plotted against increases in smartphone and social media use over the past two decades, the mental health statistics described above lay the foundation for a powerful narrative: that digital technologies are a key contributor to the deterioration of youth mental health.

3.1.2 A Brief History of Australian Mobile Phone Bans

During the 2010s, as claims about the mental health effects of smartphones and social media gained traction, often in the absence of robust empirical evidence, public discourse shifted from early optimism to growing alarm over digital connectivity (Albris et al., 2024). Governments responded with new regulations, many of them targeting adolescents' use of digital devices. By 2023, one in four countries had implemented mobile phone bans in public schools, generally with the aim of reducing distraction, improving academic outcomes, curbing cyberbullying, and promoting pro-social behaviour (UNESCO, 2023). In support of mobile phone bans, policymakers typically rely on two sources of evidence: first, findings from the psychology literature discussed above; and second, a 2016 study reporting academic gains from school bans among lower-achieving students (Beland and Murphy, 2016).

In Australia, mobile phone bans reflected community concerns: in 2019, around 80% of adults supported banning mobile phones in classrooms (Selwyn, 2019). That same year, the governments of Victoria, Western Australia and Tasmania announced bans to be implemented across public schools in Terms 1 and 2 of the 2020 school year. James Merlino, then premier of Victoria, framed the policy as a response to cyberbullying, stating that "[h]alf of all young people have experienced cyberbullying [...] by banning mobiles we can stop it at the school gate" (State Government of Victoria, 2019). In contrast, the premier of Western Australia cited improved social behaviour and psychological well-being as the primary goal, while casting doubt on the policy's ability to curb cyberbullying, which she noted mostly "happens [...] outside of school hours" (ABC News, 2019).

The bans introduced across Victoria, Western Australia and Tasmania share similar provisions. They apply only to public schools, with private institutions setting their own policies. Under the "Off and Away All Day" slogan, students must switch off phones and internet-connected devices throughout the school day, with exceptions made for medical or educational needs. Parent-student communication is routed entirely through school administrations. While Victoria and Tasmania require locked storage, Western Australia permits high-school students to carry switched-off phones as long as they remain out of sight. In 2023, New South Wales, South Australia, and

the Northern Territory adopted very similar rules, all of them requiring locked storage of phones. In 2024, Queensland and the Australian Capital Territory followed suit, making mobile phone bans effectively nationwide.

3.2 Data

My analysis combines state-level mobile phone introduction dates with longitudinal data on Australian schoolchildren and draws on supplementary sources to account for potential confounding from COVID-19.

3.2.1 Data Sources

Household, Income and Labour Dynamics in Australia (HILDA) Survey. HILDA is a nationally representative longitudinal survey, launched in 2001 and currently in its 23rd wave ([Melbourne Institute of Applied Economic and Social Research, 2024](#)). The initial sample of 7,862 households was expanded to nearly 10,000 in 2011, covering over 17,000 individuals. The survey collects detailed data on family dynamics, economic well-being, subjective well-being and labour market outcomes. Household members are interviewed from age 15 onwards, meaning that younger children are observed only indirectly through parent reports.

While geographic information is limited to state and section of state, the dataset records whether respondents are enrolled in school, which year level they are in, and the type of school they attend (government, Catholic non-government, other non-government or other). Each observation includes the exact interview date, allowing precise calculation of the time elapsed relative to the implementation of state mobile phone bans.

Mental health is assessed using two clinically validated measures: the Kessler Psychological Distress Scale (K10) and the SF-36 Mental Health score. While Kessler scores are available biennially since wave 7, the SF-36 questionnaire has been administered annually since wave 1. The aggregate SF-36 Mental Health score is derived from several survey questions, each rated on a 5- or 6-point Likert scale. These items are organised into three distinct subdomains, labelled *mental health*, *social functioning*, and *vitality*. The SF-36 manual explicitly recognises the latter two domains as integral components of mental health, broadly defined ([Ware et al., 1994](#)). The exact questions and response options are listed in Table A1 in the Appendix. Additional information on psychiatric diagnoses and take-up of prescription medication for mental health conditions is collected in waves 9, 13, 17, and 21.

HILDA does not collect information on smartphone use or screen time. While such measures would be useful for capturing variation in individual exposure, their absence

is less critical given the near-universal smartphone ownership among Australian adolescents (Rhodes, 2017).

Mobile Phone Expansion Date Data. The dates of state-level mobile phone ban introductions are collected from government press releases. Table A2 in the Appendix provides a breakdown of exact dates and sources. The introduction dates are set to the first school day on which public school students were effectively exposed to bans in their states. For ease of exposition, I occasionally refer to three treatment expansion groups, that is, groupings of states based on the year in which bans were first implemented: 2020, 2023, and 2024. While exact implementation dates vary within each group, the grouping aligns with the annual frequency of the data and does not affect the empirical strategy.

COVID Mortality Data. To account for confounding from COVID-19, I use state-level COVID-related mortality data from March 2020 to August 2023, collected from state health departments. COVID mortality reflects the intensity of the pandemic across states and over time, and served as a primary input for policy decisions on school closures and curfews (Edwards et al., 2022). It therefore provides a useful proxy for the level of disruption to students' lives during the study period. I express mortality per 100,000 residents using state-level population data from the national statistics office, as absolute numbers are only meaningful relative to population size (Australian Bureau of Statistics, 2024).

School Closure Data. To further address potential confounding from COVID-19, I collect state-level data on the exact dates of school closures due to the pandemic, using government press releases and newspaper reports. School closures occurred between 2020 and 2021, and varied across states. Within states, the duration of closures sometimes differed by region and year level. For instance, older students generally experienced less time in remote learning. To account for these two sources of intra-state variation, I use the year-level and section-of-state variables provided by the HILDA survey.

In compiling school closure timelines, I draw on data from Gillitzer and Prasad (2024), though the vast majority of my sample consists of students in higher year levels not covered in their work.¹ Table A3 in the Appendix documents the closure dates accounted for in my analysis. I further supplement this with official term dates from state education departments. Combining these sources enables me to construct a continuous treatment variable: the effective number of days a school-aged respondent was exposed to mobile phone bans at the time of the interview, accounting for closures, holidays and weekends.

¹I am very grateful to Christian Gillitzer and Nalini Prasad for sharing their data with me.

3.2.2 Summary Statistics

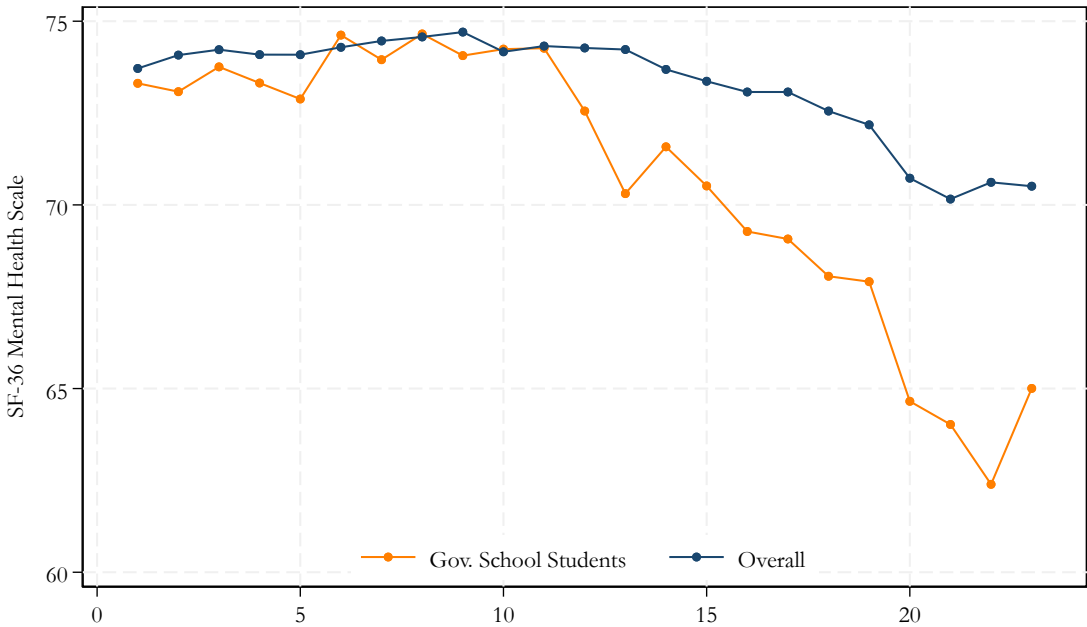
Figure A1 plots the the number of school-aged respondents to the HILDA survey over time, distinguishing between government school students and others. Respondent numbers remain broadly stable across waves, apart from notable increases in waves 11 and 23, which reflect sample top-ups. Table A4 presents summary statistics for the full sample of survey respondents enrolled in government schools at the time of the interview. Means and standard deviations are calculated from pre-treatment observations, that is, data collected prior to 2020. The sample is gender-balanced, with an average age of 16. The table also provides a range of socio-economic indicators to provide context on the population studied. It should be noted that, on average, respondents attending government schools score lower on measures such as the household's Education and Occupation Decile, the Socio-Economic Advantage Decile, and both parents' weekly earnings compared to their peers in non-government schools. This indicates that the sample analysed here is more socio-economically disadvantaged than the broader population of school-aged students. Lastly, the table presents the average SF-36 Mental Health score (72.18), alongside the individual survey items from which it is derived. As these summary statistics are based on pre-treatment data, the reported mental health levels are relatively high compared to more recent periods, a point I return to below.

Table A5 reports summary statistics by mobile phone ban expansion group, indicating that baseline differences across groups are small. The 2020 expansion group shows slightly lower scores on socio-economic indicators such as father's weekly earnings, and reports higher average home debt. Differences across all mental health survey items are minimal, suggesting comparable pre-treatment mental health across groups. Notably, baseline differences across groups are not inherently problematic in a difference-in-differences framework. Provided that groups follows parallel trends, any level differences are effectively differenced out in the estimation.

Figure A2 illustrates that the trajectory of COVID-19 mortality (expressed in deaths per 100,000 residents) across Australian states, confirming that pandemic experiences varied geographically.

Figure 3.1 below documents the trend in mental health among government school students aged 15 and above, the main sample used in this thesis. Their mental health trajectory mirrors the national and global patterns of decline discussed previously: the SF-36 Mental Health score, which ranges from 0 to 100, has fallen significantly from around 74 in wave 10 (2010) to 65 in wave 23 (2023).

Figure 3.1. Evolution of Mental Health among Australian Government School Students



Notes: This figure shows average SF-36 Mental Health scores by survey wave for government school students aged 15 and above, compared to all survey respondents. Higher scores indicate better mental health. Observations with missing or invalid responses are excluded from the averages.

Figure A3 in the Appendix extends this picture to all Australian adolescents aged 15 to 24 and plots the overall score alongside the individual items that make up the mental health subdomain. All of them exhibit clear signs of deterioration over time. The same holds for the Kessler Psychological Distress scale and its components, displayed in Figure A4.

Chapter 4

Empirical Strategy

4.1 Construction and Validation of Main Variables

My primary outcome variable is the SF-36 Mental Health score, a composite and continuous index ranging from 0 to 100, derived from the survey items described above and reported directly at the individual level in the HILDA dataset. This composite index is used in its original form in the analysis. In addition to the aggregate measure, I also analyse its individual components to provide a more granular view of mental health. This section outlines how I construct and transform these individual outcome variables. It also describes how I validate the aggregate SF-36 scale by relating it to self-reported mental healthcare diagnoses and the Kessler Psychological Distress scale.

4.1.1 Construction of Outcome Variables

Although the SF-36 Mental Health score serves as my primary outcome, I also examine individual survey items. To ensure transparency and avoid concerns about selective reporting, I include all items from the SF-36 questionnaire that contribute to the mental health summary score, as specified in the official manual (Ware et al., 1994). These items fall under the labels of *mental health*, *vitality*, and *social functioning*, and align with the definition of mental health given at the start of the previous chapter.¹

I group all survey items according to their SF-36 label – *mental health*, *vitality*, or *social functioning* – and construct three corresponding indices. Indices are built as follows: (i) I reorient all items so that higher values consistently indicate better mental health; (ii) I standardise each item using means and standard deviations from the pre-treatment period; (iii) I compute an equally weighted average of the standardised items within each group; and (iv) I standardise the resulting index. The final values can therefore be interpreted as *z*-scores, with higher values reflecting improved mental health.

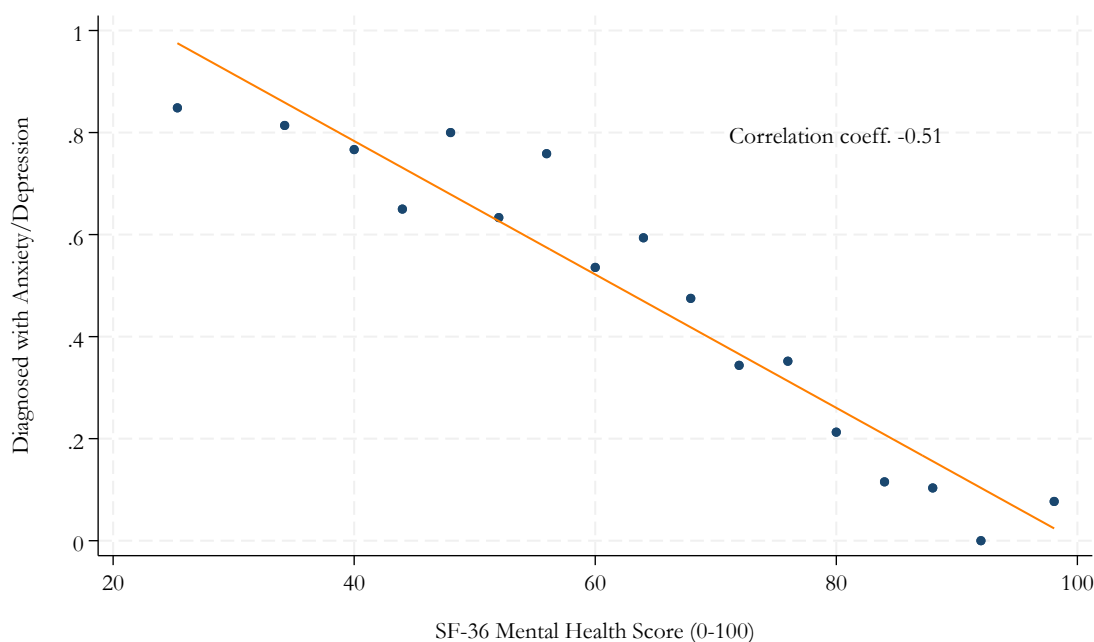
¹The relevant questions are listed in Table A1 in the Appendix.

4.1.2 Internal Validation of the Primary Outcome Variable

To demonstrate the validity of the SF-36 Mental Health scale, I draw on additional mental health related information available in the HILDA dataset. First, I examine its association with the Kessler (K10) score (Kessler et al., 2003), a widely used and clinically validated measure of psychological distress (Lawrence et al., 2015). The K10 is derived from ten questions on emotional states, each with a five-level response scale. Scores range from 10, indicating no distress, to 50, indicating severe distress. In HILDA, the K10 questionnaire is administered every second wave. Figure A5 in the Appendix shows a strong association between a high score on the K10 and a low SF-36 score, with a correlation coefficient of -0.79.

Second, Figure 4.1 relates the SF-36 score to self-reported diagnoses of depression or anxiety. The figure shows that lower SF-36 scores are strongly associated with a higher likelihood of reporting such diagnoses.

Figure 4.1. Relationship between SF-36 and Fraction Diagnosed with Depression or Anxiety



Notes: Data on anxiety and depression diagnoses is provided in waves 9, 13, 17 and 21 of the HILDA survey. The sample is restricted to respondents aged 15+ enrolled in school at the time of the interview. The horizontal axis plots the SF-36 score, with higher values indicating better mental health. The vertical axis indicates whether respondents report having been diagnosed with anxiety or depression. Each dot represents the average diagnosis rate within one of twenty quantile bins of the SF-36 distribution. A linear regression line is overlaid to highlight the negative association between SF-36 scores and the probability of diagnosis.

As a final validation exercise, I examine the relation between the SF-36 scale and self-reported visits to a mental health professional in the past 12 months. Figure A6 in the Appendix shows a robust negative relationship, with a correlation coefficient of -0.37.

Taken together, these associations lend credence to the internal consistency of the mental health data in HILDA. Higher SF-36 scores are strongly associated with worse mental health outcomes across several different metrics.

4.1.3 Main Sample and Construction of Treatment Indicator

The main sample consists of HILDA respondents who (i) report being enrolled in school at the time of the interview; and (ii) report attending a government school. Respondents enrolled in non-government schools are excluded, as no data is available on the mobile phone strategies of those schools. Within this sample, individuals are classified as treated if they were interviewed after the effective start date of mobile phone bans in their state's public schools.

4.2 Identification

4.2.1 Baseline Two-Way Fixed Effects Model

To identify the causal effect of school mobile phone bans on student mental health, I implement a staggered difference-in-differences strategy. This strategy compares the pre- and post-treatment difference in outcomes between students in states where bans were introduced and students in states that did not change their mobile phone ban status between the two periods. Since all states had adopted bans by early 2024, I exclude data from that year due to the absence of a valid control group. The last two adopting states, Queensland and the Australian Capital Territory, therefore form the non-expansion or never-treated group.

Table 4.1. Staggered Implementation of School Mobile Phone Bans across Australian States

Expansion Year	Regions	% Regions	% LGAs	% Respondents (2023)
2020	Victoria, Western Australia, Tasmania	37	46	41
2023	New South Wales, South Australia, Northern Territory	37	40	31
Non-Expansion	Australian Capital Territory, Queensland	25	14	28

Notes: Regions include all internal states and territories of Australia, except the Jervis Bay Territory, which is not covered by HILDA. LGAs refer to local government areas, the third level of government. Respondents are aged 15+, and enrolled in government schools at the time of the interview. Respondent shares are calculated from wave 23 of the survey.

Table 4.1 summarises the treatment rollout, showing that each expansion group represents a substantial share of local government areas (LGAs) and school-aged respondents in the HILDA survey.

As a first step, I estimate the staggered rollout design using a two-way fixed effects (TWFE) model, which has long been the dominant approach in applied difference-in-differences settings. Specifically, I estimate the following baseline equation:

$$Y_{ist} = \alpha_s + \delta_t + \beta \cdot \text{PhoneBan}_{st} + \gamma \cdot \mathbf{X}_{it} + \psi \cdot \mathbf{X}_{st} + \varepsilon_{ist} \quad (4.1)$$

where Y_{ist} represents a mental health outcome of individual i , who resides in state s , interviewed on date t ; α_s denotes state fixed effects;² δ_t denotes survey-wave (that is, year) fixed effects; PhoneBan_{st} is an indicator set to one if a mobile phone ban was in effect in state s at date t ; $\mathbf{X}_{it}$ and $\mathbf{X}_{st}$ are vectors of individual-level and state-level controls. I estimate equation 4.1 using OLS. Following [Abadie et al. \(2023\)](#), I cluster standard errors at the level of treatment assignment, that is, the state level.

The coefficient of interest β identifies the average treatment effect on the treated (ATT) of the introduction of mobile phone bans on student mental health under several assumptions. First, the *No Anticipation* assumption requires that individuals do not alter their behaviour in anticipation of future treatment. Second, the *Parallel Trends* assumption requires that, in the absence of treatment, outcomes would have evolved along the same trajectories across groups. In staggered designs, this assumption can take different forms depending on whether the control group consists of the never-treated, the not-yet-treated, or both; I use both. Finally, recent work has shown that TWFE models yield unbiased estimates of the ATT only under an additional condition: that treatment effects be homogenous across groups and over time ([De Chaisemartin and d'Haultfoeuille, 2020](#)). I address this limitation below.

Under the previously stated assumptions, the TWFE model addresses several potential threats to causal inference. First, state fixed effects control for time-invariant differences in student mental health across states. For example, persistent differences in wealth, health infrastructure, or educational institutions may lead to better (or worse) student mental health outcomes. Second, survey-wave fixed effects account for nationwide shocks or trends that affect all states similarly. For example, macroeconomic fluctuations or national policy changes may influence household income and, in turn, student mental health.

The plausibility of the parallel trends assumption may be doubted in this setting. States might have introduced mobile phone bans at different times precisely because they were on diverging mental health trajectories. In particular, bans may have been im-

²I also provide an alternative specification with individual-level fixed effects.

plemented in response to deteriorating student mental health, implying that treatment timing is correlated to underlying trends in the outcome. I address this concern in three ways. First, I estimate event study versions of equation 4.1 to test for the presence of differential pre-treatment trends. Specifically, I estimate the following model:

$$Y_{ist} = \alpha_s + \delta_t + \beta_k \cdot \sum_k^T D_{k(st)} + \gamma \cdot \mathbf{X}_{it} + \psi \cdot \mathbf{X}_{st} + \varepsilon_{ist} \quad (4.2)$$

where $D_{k(st)}$ is a set of dummy variables taking the value one if, for state s in survey-wave t , the introduction of mobile phone bans was k years away. This specification also captures dynamic treatment effects, a subject to which I return below. Second, I re-estimate equation 4.2 with a range of robust event study estimators and formally test whether pre-treatment effects (that is, placebo leads) are statistically significant (De Chaisemartin and d’Haultfoeulle, 2020; Sun and Abraham, 2021; Borusyak et al., 2024). Third, most simply, visual inspection of student mental health trends by treatment expansion group, shown in Figure A7 in the Appendix, reveals no evidence that differential treatment timing coincides with diverging pre-treatment mental health trajectories.

4.2.2 Accounting for Pandemic-Related Confounding

A second threat to the parallel trends assumption arises from the COVID-19 pandemic. The first confirmed case in Australia occurred on January 25, 2020 in Victoria, just days before the first mobile phone bans were implemented. Given its well-documented impact on adolescent mental health, and the disparities in numbers of cases, numbers of deaths, and related disruption across states, COVID-19 may have induced differential post-treatment trends in mental health outcomes.

To address this, I adopt a three-pronged strategy. First, I control for individuals’ self-reported level of disruption due to the pandemic, based on HILDA survey items in waves 20 to 23, which ask how much respondents’ lives changed due to COVID and the extent to which they followed social distancing rules. Second, I include biweekly state-level COVID mortality rates (in deaths per 100,000), which proxy the intensity of the pandemic’s impact and public health responses. Third, I re-estimate results using a continuous treatment variable capturing the number of days students were exposed to mobile phone bans when interviewed, accounting for COVID-related school closures.

When control variables are included, the parallel trends assumption becomes conditional on covariates, meaning that groups can experience differential trends, so long as these are fully explained by changes in observed controls (Baker et al., 2025). A recent working paper by Caetano et al. (2022) highlights challenges with difference-

in-difference designs that incorporate time-varying covariates, particularly the risk of introducing "bad controls", that is, covariates that are affected by the treatment. In my setting, this is unlikely to pose a problem, as mobile phone bans presumably did not influence COVID mortality rates. However, TWFE estimates remain sensitive to functional form, even when researchers are confident that time-varying controls evolve exogenously from the treatment.

To complement the TWFE estimates, I implement the event study estimator proposed by [De Chaisemartin and d'Haultfoeuille \(2024\)](#), which accommodates time-varying controls. This approach allows counterfactual post-treatment mental health trends to vary across states. Provided these differential trends are captured by a linear model in the change in COVID mortality and other controls such as self-reported pandemic-related disruption, the method yields unbiased estimates of the ATT.

4.2.3 TWFE Limitations and Alternative Estimators

Recent econometric research has highlighted serious limitations of extending TWFE estimators beyond the classical difference-in-differences setting with two groups and two periods. The core issue is that TWFE implicitly uses already-treated units as controls. As shown by [Goodman-Bacon \(2021\)](#), the TWFE estimator in staggered adoption designs can be decomposed into a weighted average of all possible 2×2 comparisons between groups that switch treatment status and groups that do not. Many of these comparisons involve already-treated units as controls, and may impose negative weights on certain ATT estimates. [Baker et al. \(2022\)](#) demonstrate that this can bias the TWFE estimate away from the true ATT, potentially even reversing the sign. A causal interpretation of the estimate is possible only under strong additional assumptions: namely, that treatment effects are constant over time and across groups ([Sun and Abraham, 2021](#)).

In my setting, these additional assumptions are unlikely to hold. First, treatment effects likely vary over time due to pandemic-related school closures and hybrid learning arrangements. Longer exposure to mobile phone bans in later years (2022–2023), when in-person schooling was more consistent, may have led to stronger mental health impacts. Second, treatment effects may differ across states. School closures varied in duration across locations, influencing how long students were exposed to bans. Moreover, policy provisions differed in one state: Western Australia did not require phones to be placed in locked storage, potentially reducing compliance and weakening the policy's mental health effects.

To account for treatment effect heterogeneity, I implement the robust event study estimator proposed by [De Chaisemartin and d'Haultfoeuille \(2024\)](#). This estimator avoids invalid 2×2 comparisons: groups that switch treatment status between pe-

riod t and $t + 1$ are compared to groups with the same treatment status in t that do not switch in $t + 1$. In other words, only two types of comparisons are permitted: (i) between groups that switch into treatment and groups that remain untreated, and (ii) between groups that switch out of treatment and groups that remain treated. In my setting, only the first type of comparison arises, since no Australian state switched out of mobile phone bans during the sample period. As a result, the difference-in-differences estimates are based exclusively on what [De Chaisemartin and d'Haultfoeuille](#) refer to as "switchers into treatment". In my staggered adoption design with multiple periods, the estimator computes a separate difference-in-differences estimate for each pair of consecutive periods for which a comparison of type (i) is possible. These estimates are then averaged across periods, delivering an unbiased estimate of the average treatment effect on the treated across all two-period switching comparisons. This procedure allows for fully heterogeneous treatment effects across time and groups, under standard no-anticipation and conditional parallel trends assumptions.

The estimator also addresses a further challenge in event study inference: since many treatment effects are estimated simultaneously, the issue of multiple hypothesis testing arises. Asymptotically correct inference requires adjusting critical values and confidence intervals to account for this. [De Chaisemartin and d'Haultfoeuille](#)'s estimator returns effects by event-time with appropriately adjusted confidence intervals.

Finally, the procedure yields an *average total effect per unit of treatment* parameter, defined as a weighted sum of period-specific treatment effects. The horizons over which the total effect is cumulated depend on the design; in my case, they span up to three post-treatment periods for the 2020 expansion group. This parameter facilitates cost-benefit analysis by quantifying the total impact of treatment relative to a counterfactual in which no treatment occurred.

Chapter 5

Results

In this section, I report estimates from a TWFE model and supplementary analyses on individual SF-36 items, before turning to an event study specification. First, the baseline TWFE estimates suggest that the introduction of mobile phone bans did not lead to a statistically significant change in overall student mental health scores. As a robustness check, I replace the binary treatment indicator with a continuous measure capturing the total number of days a student was subject to mobile phone bans. This alternative specification also yields no evidence of an effect. Next, I examine each SF-36 item underlying the summary measure. Consistent with the aggregate result, none of the individual items exhibits a significant shift at conventional levels. I then repeat the analysis separately for male and female students. In the male subsample, bans lead to a statistically significant reduction in the frequency with which emotional or physical health interfered with social activities. This result remains after adjustment for multiple hypothesis testing. By contrast, none of the estimates for female students attain significance. Although most point estimates suggest slightly more favourable effects among boys than girls, the absence of statistical significance, which is further exacerbated by adjustments for simultaneous inference, precludes a systematic interpretation of sex differences.

Finally, I implement event study estimators to test for divergent pre-treatment trends in mental health outcomes and to explore dynamic effects. These estimates support the parallel trends assumption and provide modest evidence of a gradually increasing treatment effect in the post-ban periods. The cumulative impact over three years, calculated following [De Chaisemartin and d’Haultfoeuille \(2024\)](#), is positive and reaches statistical significance, though the magnitude is small in absolute terms.

5.1 TWFE Baseline Results

5.1.1 Effects on the Aggregate Mental Health Index

Table 5.1 reports estimates of the effect of mobile phone bans on student mental health, measured via the SF-36 summary mental health measure (0-100, with higher values indicating better outcomes; standard deviation ≈ 17). All specifications rely on a binary treatment indicator, rather than a continuous measure of exposure.

Column (1) presents a basic specification with only state and survey-wave fixed effects. The point estimate is small, negative and far from significant ($p > 0.5$). This specification omits any controls for individual characteristics or COVID-related shocks, which may confound the relationship between phone bans and mental health. Given the uneven spread of the pandemic across Australia and the heterogeneity in local policy responses, the parallel trends assumption is likely violated in this setup.

Table 5.1. TWFE Estimates of the Effect of Mobile Phone Bans on Student Mental Health

	SF-36 Mental Health Scale (0-100)		
	(1)	(2)	(3)
<i>Expl. Variables:</i>			
Under Treatment	-0.9742 (1.687)	0.0806 (1.237)	-1.368 (2.394)
Observations	9,028	9,024	9,027
R ²	0.04105	0.11237	0.75356
Controls		✓	✓
State fixed effects	✓	✓	
Year fixed effects	✓	✓	✓
Individual fixed effects			✓

Notes: All models use a binary treatment indicator for exposure to mobile phone bans. Standard errors are clustered at the state level. Control variables include student sex, household structure, section of state, long-term health conditions, self-reported pandemic-related disruption, and state-level COVID-19 mortality.

Column (2) introduces a range of control variables. First, it accounts for time-invariant student characteristics, namely, sex, section of state, household structure, and potential long-term health conditions. Their inclusion strengthens the case for the parallel trends assumption: it now needs to hold only within each covariate-specific stratum rather than across the whole population. More importantly, this specification adds

time-varying covariates: students' self-reported level of disruption from the pandemic and biweekly state-level COVID-related mortality. These adjustments account for differential exposure to the pandemic, a source of potential confounding. This is my preferred specification, offering a credible claim to identification under conditional parallel trends. The estimated treatment effect turns slightly positive, but remains very small and statistically insignificant.

Column (3) replaces state fixed effects with individual fixed effects, absorbing all variation in mental health due to time-invariant student characteristics. The estimate shifts back to negative but remains statistically indistinguishable from zero. This specification may over-correct: since most students enter the panel at age 15, and are observed only over few periods before they leave school, fixed effects are estimated from few data points. Where students are observed only after the implementation of bans, fixed effects may absorb some of the relevant variation caused by the treatment.

Across all models, there is no evidence that mobile phone bans significantly affected students' mental health. Several explanations may account for this null result. First, reduced smartphone use may simply not improve mental health outcomes among Australian schoolchildren aged 15 and above. This leaves open the possibility that they affect the mental health of younger adolescents. Second, students may have offset bans by increasing use outside school hours, leaving total exposure unchanged. If so, any potential positive effects of a reduction in net exposure would not be captured here. Third, treatment effects may differ across subgroups and cancel each other out in aggregate. For instance, some students may benefit from online support networks while others may suffer from cyberbullying, such that analysis at the population level masks important heterogeneity. Fourth, effects may be dynamic or delayed, especially given COVID-related disruptions to schooling, which reduced the time students spent in school during the early years of bans.

Table A6 in the Appendix reports parallel TWFE results using a continuous treatment variable instead of a binary indicator. The continuous variable captures the total number of days students were exposed to mobile phone bans, accounting for COVID-related school closures and holidays. While this approach adjusts for variation in treatment duration, it does not fully address COVID-related confounding, as the pandemic likely affected mental through channels unrelated to school closures. I therefore retain COVID mortality and self-reported pandemic-related disruption as controls. The results remain unchanged: across specifications, there is no evidence of significant treatment effects.

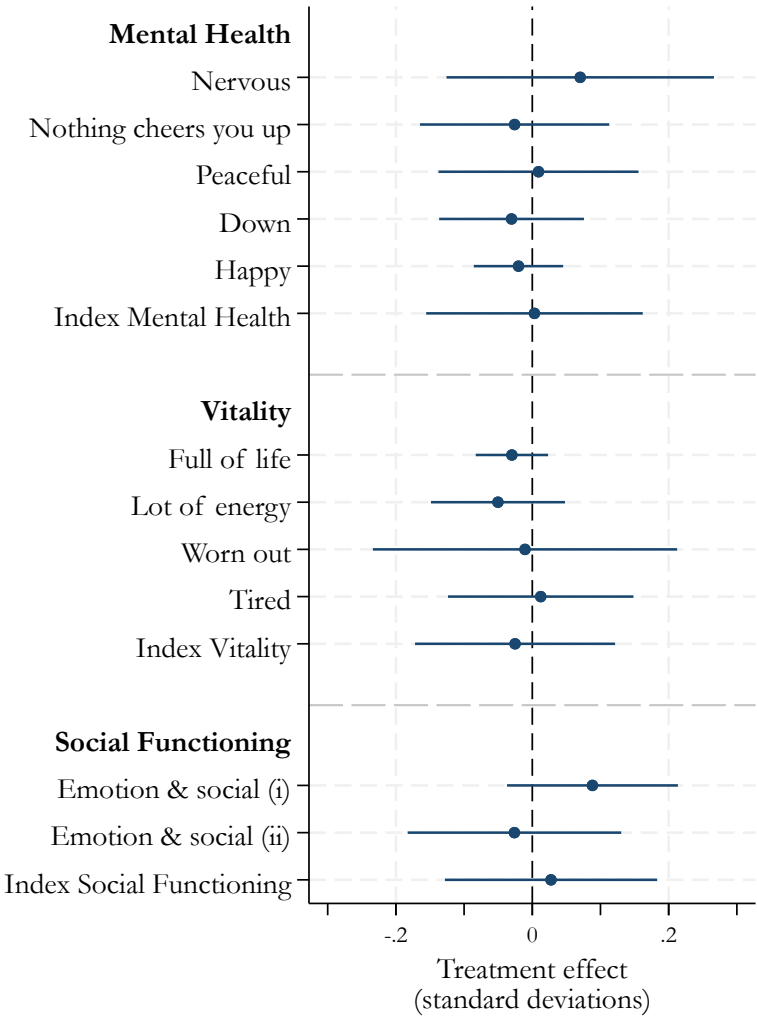
It should be noted that finding no effect of mobile phone bans on student mental health does not in itself constitute evidence against the view that rising levels of smartphone adoption contributed to the long-term deterioration of adolescent mental health. Mo-

bile connectivity may well have had deep, cumulative effects on adolescents’ time allocation, habits, aspirations, and even cognitive processes. These shifts may have become entrenched and gradually eroded mental health over time. Yet there is no guarantee that an abrupt halt to smartphone exposure at school simply reverses those changes or restores some earlier equilibrium.

5.1.2 Disaggregated Effects on Individual Survey Items

Figure 5.1 presents estimated effects of mobile phone bans on the individual SF-36 items that make up the summary mental health measure. All regressions use a binary treatment indicator and include state and survey-wave fixed effects, along with the full set of control variables discussed earlier.

Figure 5.1. Treatment Effects on Mental Health, Vitality, and Social-Functioning Items



Notes: Coefficients show estimated treatment effects of mobile phone bans on standardized SF-36 mental health items. Error bars represent 95% confidence intervals. Estimates derive from TWFE regressions with survey-wave and state fixed effects, as well as controls for student-level characteristics and state-level COVID mortality. Standard errors are clustered at the state level.

The figure organizes items into nested families, *mental health*, *vitality*, and *social functioning*, as defined by the SF-36 framework. Exact question wording is reported in Table A1 in the Appendix. The final coefficient in each group corresponds to an equally weighted index of the preceding items.

Consistent with the aggregate results, none of the item-level estimates is statistically significant. Most points estimates are close to zero and vary in sign. These findings are in line with those reported in the psychology literature, which often finds small effects of either sign (Odgers and Jensen, 2020).

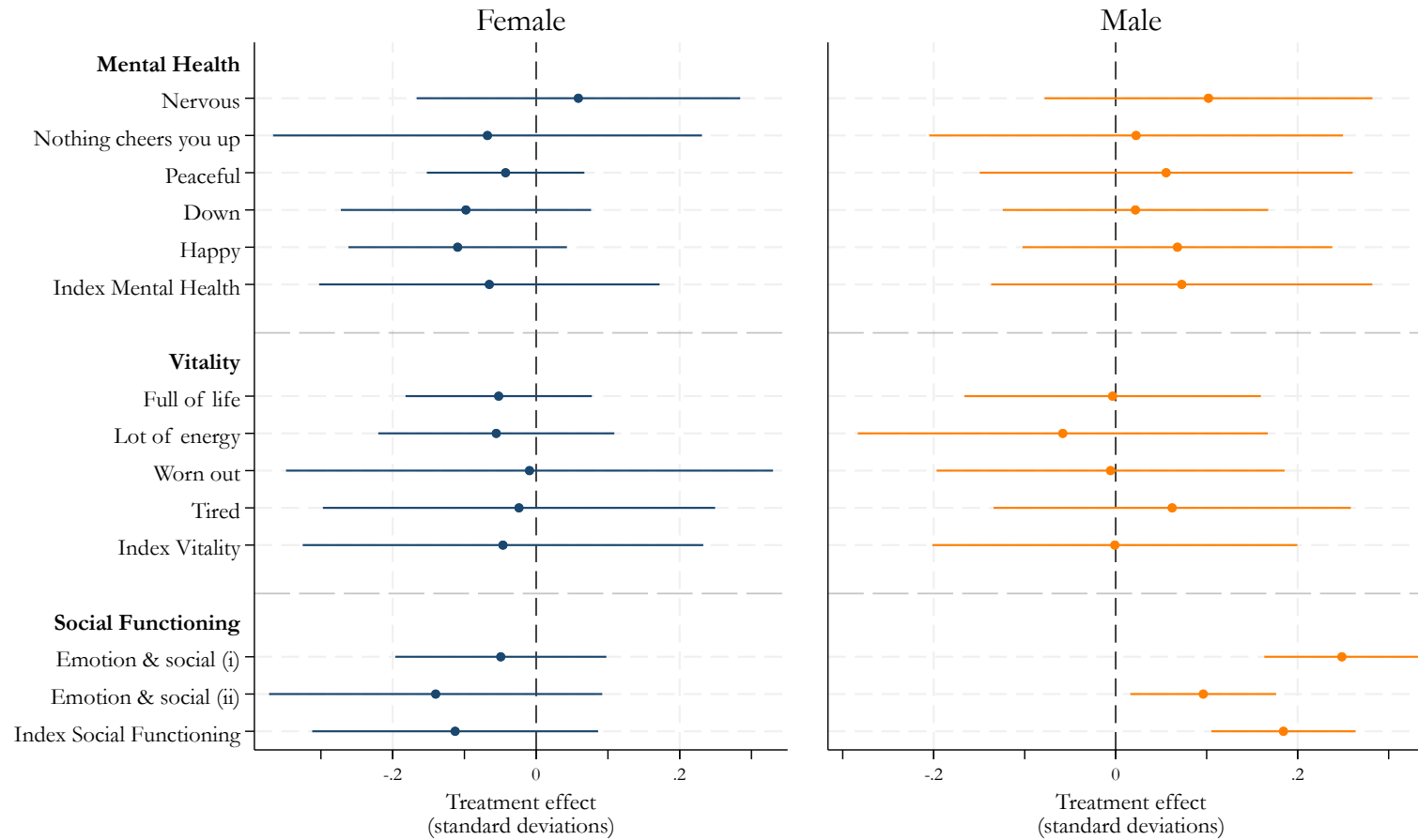
It should be noted that estimating multiple regressions across different outcome variables mechanically increases the false discovery rate (FDR) due to multiple hypothesis testing. However, since none of the coefficients is significant, there is no need to adjust for the FDR at this stage. In the next section, such corrections are applied.

5.1.3 Heterogeneity

Figure 5.2 replicates the same regressions separately for male and female students. The pattern of results shifts somewhat: point estimates indicate more favourable effects for males than for females. Most effects remain statistically indistinguishable from zero, with the exception of survey items capturing social functioning among male students. Specifically, coefficients suggest that mobile phone bans led male students to report that emotional or physical health interfered (i) less frequently and (ii) to a lesser extent with normal social activities. The effect is highly significant for the frequency item and somewhat less so for the extent item, yielding an effect on the composite social functioning index of 0.18 standard deviations. To contextualise, the raw survey items on social functioning are assessed on a 5-point scale, with a standard deviation of around 0.9 in the pre-treatment period (see Table A4). This implies that the point estimate corresponds to a relatively small shift in reported outcomes.

To account for multiple testing, I control the false discovery rate (FDR), that is, the expected proportion of false positives among the total number of rejected null hypotheses (or equivalently, the type I error rate). Benjamini et al. (2006) provide a procedure that allows to control the FDR at a desired level q . Building on this work, the algorithm by Anderson (2008) reports the smallest level of the FDR at which a given null hypothesis would be rejected. These "sharpened q -values" serve as a multiple-testing-adjusted analogue to standard p -values.

Figure 5.2. Treatment Effects on Mental Health, Vitality, and Social-Functioning Items by Sex



Notes: Coefficients show estimated treatment effects of mobile phone bans on standardized SF-36 mental health items, reported separately for male and female students. Error bars indicate 95% confidence intervals. Estimates are obtained from TWFE regressions with survey wave and state fixed effects, and include controls for student-level characteristics and state-level COVID mortality. Standard errors are clustered at the state level.

Most outcomes yield sharpened q -values of 1, implying that they would be statistically significant only with an FDR of 100%. Such high q -values are common when making simultaneous inferences across many outcomes (Anderson, 2008, p. 1492). In contrast, the first social functioning item yields a sharpened q -value of 0.004 ($p = 0.00024$), and the second a q -value of 0.087 ($p = 0.01999$). The composite social functioning index has a q -value of 0.006 ($p = 0.00087$). Full results for the male subsample, including multiple testing corrections, are reported in Table A7 in the Appendix. Taken together, these results provide evidence that mobile phone bans significantly improved the social functioning of male students, even after adjusting for multiple inference.

The absence of detectable effects among female students may stem from distinct smartphone usage patterns. Research consistently finds that girls are more likely to use smartphones for social networking, while boys allocate a greater share of screen time to electronic gaming (Claesdotter-Knutsson et al., 2021). In Australia, a 2017 Royal Children's Hospital poll reported that among teenagers aged 13 to 18, boys were twice as likely as girls to engage in electronic gaming, while girls were twice as likely to use social media (Rhodes, 2017). During the pandemic, these patterns intensified: increases in screen time were primarily driven by gaming for boys and social networking for girls (Joshi and Hinkley, 2021).

Although speculative, these differences suggest a mechanism behind the observed heterogeneity. If girls used phones during school hours primarily for online social interaction, then the bans may have changed the mode, but not the amount, of social engagement, merely shifting it from online to offline. By contrast, if boys used phones more for non-social activities such as gaming, the bans may have led them to reallocate time to social interaction, displacing non-social screen time. This reallocation could have improved boys' social skills and lowered emotional barriers to social activity, contributing to the gains observed in measures of social functioning.

Anecdotal evidence supports the idea that phone bans increased peer interaction. One principal participating in a 2019 pilot phone ban reported to the Western Australian Premier that they "hadn't anticipated the level of noise in the playground at lunchtime because kids were actually talking to each other" (ABC News, 2019).

Finally, because bans disrupted a form of communication more central to girls than boys, girls may have experienced greater adjustment costs. These may have offset potential gains from increased face-to-face interaction, resulting in null effects on measures of social functioning. U.S. survey data lends credence to this hypothesis: in 2022, 58% of teen girls, versus 49% of boys, reported it would be difficult to give up social media, with only 15% of girls saying it would be very easy (Vogels et al., 2022).

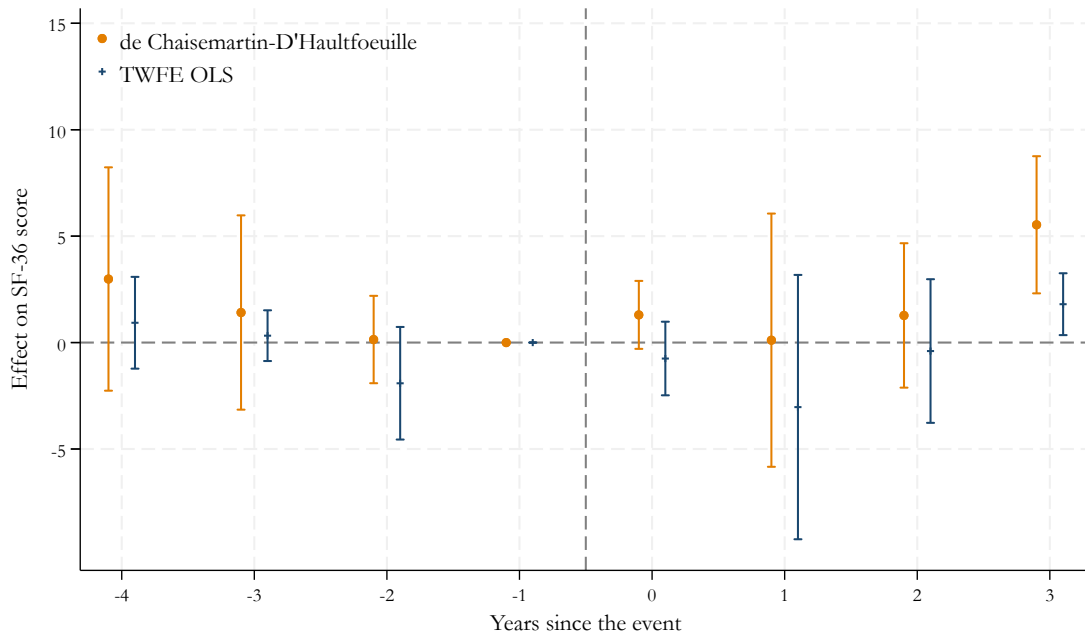
5.2 Event Study Results

This section presents event study estimates of the effect of mobile phone bans on student mental health. The first subsection assesses pre-treatment trends and examines the dynamic evolution of treatment effects over time. The second subsection introduces a summary measure of the cumulative treatment effect and provides an interpretation of its magnitude.

5.2.1 Baseline Findings

Figure 5.3 presents event study estimates of the effect of mobile phone bans on student mental health, using both standard TWFE and the robust estimator proposed by De Chaisemartin and d'Haultfoeuille (2024). The year prior to treatment implementation serves as the reference period, with the corresponding coefficient being normalized to zero. As before, a binary treatment indicator is used.

Figure 5.3. Event Study Estimates of Mobile Phone Ban Effects on Student Mental Health



Notes: The figure plots estimated effects of mobile phone bans on the SF-36 mental health score, using two event study estimators: a conventional TWFE and the estimator proposed by De Chaisemartin and d'Haultfoeuille (2024). Estimates are relative to the period immediately preceding treatment. All estimations control for student-level covariates, survey-wave and state fixed effects, and state-level COVID mortality. Standard errors are clustered at the state level. Vertical lines indicate 95% confidence intervals.

These estimates serve two purposes. First, they allow a visual and statistical assessment of potential pre-treatment trends in mental health outcomes. Under no anticipation and parallel trends, pre-treatment coefficients, generally referred to as "placebos", equal zero in expectation.

Across both estimators, all pre-treatment coefficients are insignificant, suggesting no differential trends prior to policy adoption. Appendix Table A8 provides individual placebo estimates and a joint test of pre-treatment effects, as recommended by De Chaisemartin and d'Haultfoeuille (2024). The joint test yields a p -value of 0.435, providing no evidence against the parallel trends assumption.

Second, event study estimates shed light on the evolution of treatment effects over time. Both estimators show post-treatment effects increasing dynamically, with the largest and only statistically significant impact occurring in the third year after implementation. Aside from this estimate, post-treatment coefficients are not significant at the 5% level. Notably, TWFE results exhibit a flatter trend in the post-period than the robust estimator, likely due to the negative weights issue discussed in Chapter 4.

Several factors may explain the increase in treatment effects over time. First, early post-treatment years coincided with the COVID-19 pandemic for the 2020 expansion group, which likely limited students' exposure to mobile phone bans due to widespread remote and hybrid learning. In contrast, the third-year effect, which is driven exclusively by the 2020 expansion group, reflects a period free from pandemic-related school disruptions. In other words, treatment effects intensify over time simply because treatment intensity increased for the largest share of treated students in the sample. Second, the effects of mobile phone bans may have emerged gradually, as students adjusted to the new school environment. Presumably, many had developed habits around uninterrupted mobile connectivity, and the potential benefits of reduced screen time, such as increased pro-social interaction or physical activity, may have taken time to materialise.

The point estimate for the third post-treatment period, as reported by the robust estimator, is sizeable, lying just over 5 points on the SF-36 scale. However, wide confidence intervals around the estimate indicate substantial statistical uncertainty. Alternative robust estimators, shown in Figure A8 in the Appendix, yield smaller but still positive and statistically significant effects (Sun and Abraham, 2021; Borusyak et al., 2024). Importantly, these estimators also reinforce the case for parallel trends: pre-treatment coefficients remain statistically insignificant, and show no evidence of systematic divergence prior to treatment.

5.2.2 Average Cumulative Effect and Magnitude Interpretation

To provide a policy-relevant measure of the overall impact of mobile phone bans on student mental health, Table 5.2 reports the average cumulative effect per treated unit. This figure aggregates the dynamic post-treatment effects estimated using the method of De Chaisemartin and d'Haultfoeuille. It also helps resolve the interpretive challenges of event study coefficients in settings with multiple groups and staggered treat-

ment timing (ibid., 2023). The average exposure period over which the total effect is accumulated is 2.2 years, reflecting the typical duration that treated students in the sample were subject to mobile phone bans. The point estimate indicates a 2.1 increase in the SF-36 Mental Health measure, and is statistically significant at the 5% level.

Table 5.2. *Average Cumulative Treatment Effect*

	Estimate	Std. Error	95% CI	N
Average cumulative effect	2.1244	.8671	[0.4249, 3.8239]	1447

Notes: This table reports the estimated average cumulative effect per unit of treatment, as reported by De Chaisemartin and d’Haultfoeuille (2024) robust procedure. The average number of time periods over which a treatment’s effect is accumulated is 2.2 years.

To contextualise the magnitude of the estimated effect, I relate it to the probability of a clinical diagnosis of anxiety or depression. Using data from adolescents aged 15 to 24 in HILDA waves 9, 13, 17, and 21, I model the probability of diagnosis as a function of the SF-36 Mental Health score using Probit regression (see Appendix Table A9). I then compute the average marginal effect of a unit increase in the SF-36 score, reported in Table A10. This average effect corresponds to the mean of all marginal effects, obtained by taking the derivative of the predicted probability function with respect to the SF-36 score across its distribution.¹ The results show that a one point increase in the SF-36 score reduces the probability of diagnosis by 0.011 on average. A simple back-of-the-envelope calculation then suggests that the average cumulative effect of mobile phone bans (that is, an increase of 2.1 on the SF-36 scale over 2.2 years), translates into a 2.3 percentage point reduction in the probability of being clinically diagnosed with anxiety or depression.

These findings of modest mental health improvements should be interpreted with caution. First, baseline TWFE estimates generally indicate no effect. Second, while alternative robust estimators point in the same direction as that of De Chaisemartin and d’Haultfoeuille, they yield somewhat smaller effect sizes. The cumulative impact reported here may therefore overstate treatment effects. Finally, the average cumulative effect is largely driven by the estimate for the third post-treatment year. While there are plausible explanations for the delayed emergence of effects, these results warrant further scrutiny and robustness checks.

¹Figure A9 in the Appendix visualizes the predicted probabilities across the full range of the SF-36 scale.

Chapter 6

Conclusion

Today's adolescents are the first generation to grow up with constant access to the internet and digital media. More than 95% of 15- to 24-year-olds in high- and upper-middle income countries use the internet, and well over 90% own a smartphone ([International Telecommunication Union, 2023](#)). The consequences of this digital transformation are likely profound and complex, ranging from changing patterns of social interaction to the emergence of new forms of entertainment, such as streaming content and online gaming. These shifts have not occurred without raising concerns from scholars, policymakers and civil society. However, the empirical evidence underpinning these concerns remains inadequate.

Focusing on a specific type of policy intervention which has grown increasingly popular in recent years, this thesis aimed to contribute well-identified causal evidence to the public debate. The analysis provides some evidence that school mobile phone bans yield mental health improvements. However, these improvements are small in magnitude and arise only in the later years of bans. The limited size of effects may reflect the scope of the policy intervention under consideration. Phone bans at school may have left usage at home unchanged, doing little to alter habits such as late-night social media use, which is known to interfere with sleep ([Hale and Guan, 2015](#)). In fact, bans may even have led to increases in phone usage outside of school hours, a possibility that complicates the interpretation of the presented findings.

The delayed emergence of positive effects, on the other hand, may indicate an adjustment process. Treated individuals were generally born after 2005, and therefore grew up with internet-connected devices from a very young age. As a result, digital habits may have reshaped their behavioural patterns in ways that are deep and not easily reversible. Undoing these developments plausibly takes time, and may even involve transitory adjustment costs. In the early years of bans, these costs may offset

the gains that might arise from, for example, a reallocation of time towards mental health-enhancing activities, such as face-to-face interactions.

Beyond the summary mental health measure, baseline two-way fixed effects regressions provide evidence of improvements in the social functioning of male students. That similar effects are not detectable among female students may, speculatively, be attributed to different baseline patterns of smartphone use, with girls more likely to use their phones for social activities. At first glance, this heterogeneity appears at odds with the existing literature, which tends to find more positive effects among women than among men (see, for instance, [Golin, 2022](#)). However, little research has focused specifically on digital media's impact on social functioning, a distinctive dimension of mental health. This suggests a fruitful avenue for future research.

Mobile phone bans were prompted in part by what may be the gravest accusation directed at digital media, or at least the one that has caused the greatest agitation: that it is a major contributing factor to recent declines in adolescent mental health. The findings presented here do not directly speak to this debate. The effects of an abrupt halt to digital media exposure need not be symmetric to those of a gradual increase. However, this thesis addresses a question of direct relevance to policymakers: given robust evidence of declines in adolescent mental health, as well as emerging evidence on negative mental health effects of the internet and social media, can mobile phone bans improve student welfare? While highlighting the need for further research, particularly outside the Australian context, and involving a broader population of school students, the findings presented here do suggest that mobile phone bans can go some way towards improving student mental health.

References

- Abadie, A., Athey, S., Imbens, G. W., and Wooldridge, J. M. (2023). When should you adjust standard errors for clustering? *The Quarterly Journal of Economics*, 138(1):1–35.
- ABC News (2019). Mobile phones to be banned in WA public schools from 2020 under McGowan Government. Published 30 October 2019. Accessed 20 May 2025.
- ACT Government (2023). New Mobile Phone Policy for ACT Public Schools. <https://www.act.gov.au/our-canberra/latest-news/2023/december/new-mobile-phone-policy-for-act-public-schools>. Accessed 2025-05-17.
- Albris, K., Fast, K., Karlsen, F., Kaun, A., Lomborg, S., and Syvertsen, T. (2024). *The digital backlash and the paradoxes of disconnection*. Nordicom, University of Gothenburg.
- Allcott, H., Braghieri, L., Eichmeyer, S., and Gentzkow, M. (2020). The welfare effects of social media. *American Economic Review*, 110(3):629–676.
- Allcott, H., Gentzkow, M., and Song, L. (2022). Digital addiction. *American Economic Review*, 112(7):2424–2463.
- Allcott, H., Gentzkow, M., Wittenbrink, B., Cisneros, J. C., Crespo-Tenorio, A., Dimmery, D., Freelon, D., González-Bailón, S., Guess, A. M., Kim, Y. M., et al. (2025). The effect of deactivating facebook and instagram on users’ emotional state. Technical report, National Bureau of Economic Research.
- American Psychological Association (2020). Mental health. <https://dictionary.apa.org/mental-health>. Retrieved from <https://dictionary.apa.org/mental-health>.
- Anderson, M. L. (2008). Multiple inference and gender differences in the effects of early intervention: A reevaluation of the Abecedarian, Perry Preschool, and Early Training Projects. *Journal of the American statistical Association*, 103(484):1481–1495.

- Antoninis, M., Alcott, B., Al Hadheri, S., April, D., Fouad Barakat, B., Barrios Rivera, M., Baskakova, Y., Barry, M., Bekkouche, Y., Caro Vasquez, D., et al. (2023). Global Education Monitoring Report 2023: Technology in education: A tool on whose terms?
- Australian Bureau of Statistics (2022). National study of mental health and wellbeing.
- Australian Bureau of Statistics (2024). National, state and territory population. <https://www.abs.gov.au/statistics/people/population/national-state-and-territory-population/sep-2024>. Accessed: 2025-05-21.
- Bailen, N. H., Green, L. M., and Thompson, R. J. (2019). Understanding emotion in adolescents: A review of emotional frequency, intensity, instability, and clarity. *Emotion Review*, 11(1):63–73.
- Baker, A., Callaway, B., Cunningham, S., Goodman-Bacon, A., and Sant’Anna, P. H. (2025). Difference-in-differences designs: A practitioner’s guide. *arXiv preprint arXiv:2503.13323*.
- Baker, A. C., Larcker, D. F., and Wang, C. C. (2022). How much should we trust staggered difference-in-differences estimates? *Journal of Financial Economics*, 144(2):370–395.
- Beland, L.-P. and Murphy, R. (2016). Ill communication: Technology, Distraction & Student Performance. *Labour Economics*, 41:61–76.
- Benjamini, Y., Krieger, A. M., and Yekutieli, D. (2006). Adaptive linear step-up procedures that control the false discovery rate. *Biometrika*, 93(3):491–507.
- Best, P., Manktelow, R., and Taylor, B. (2014). Online communication, social media and adolescent wellbeing: A systematic narrative review. *Children and Youth Services Review*, 41:27–36.
- Borusyak, K., Jaravel, X., and Spiess, J. (2024). Revisiting event-study designs: robust and efficient estimation. *Review of Economic Studies*, 91(6):3253–3285.
- Braghieri, L., Levy, R., and Makarin, A. (2022). Social media and mental health. *American Economic Review*, 112(11):3660–3693.
- Caetano, C., Callaway, B., Payne, S., and Rodrigues, H. S. (2022). Difference in differences with time-varying covariates. *arXiv preprint arXiv:2202.02903*.
- Claesdotter-Knutsson, E., André, F., Fridh, M., Delfin, C., Hakansson, A., and Lindström, M. (2021). Gender-based differences and associated factors surrounding excessive smartphone use among adolescents: Cross-sectional study. *JMIR pediatrics and parenting*, 4(4):e30889.

- Collishaw, S. (2015). Annual research review: secular trends in child and adolescent mental health. *Journal of Child Psychology and Psychiatry*, 56(3):370–393.
- Collishaw, S., Maughan, B., Natarajan, L., and Pickles, A. (2010). Trends in adolescent emotional problems in England: a comparison of two national cohorts twenty years apart. *Journal of Child Psychology and Psychiatry*, 51(8):885–894.
- Connell, J. P. and Wellborn, J. G. (1991). Competence, autonomy, and relatedness: A motivational analysis of self-system processes.
- Copeland, W. E., Angold, A., Shanahan, L., and Costello, E. J. (2014). Longitudinal patterns of anxiety from childhood to adulthood: the great smoky mountains study. *Journal of the American Academy of Child & Adolescent Psychiatry*, 53(1):21–33.
- Daly, M. (2018). Social media use may explain little of the recent rise in depressive symptoms among adolescent girls. *Clinical Psychological Science*, 6(3):295–295.
- De Chaisemartin, C. and d’Haultfoeuille, X. (2024). Difference-in-differences estimators of intertemporal treatment effects. *Review of Economics and Statistics*, pages 1–45.
- De Chaisemartin, C. and d’Haultfoeuille, X. (2020). Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, 110(9):2964–2996.
- De Chaisemartin, C. and d’Haultfoeuille, X. (2023). Two-way fixed effects and differences-in-differences with heterogeneous treatment effects: A survey. *The Econometrics Journal*, 26(3):C1–C30.
- Department for Education, South Australia (2023). Mobile phones and personal devices in public schools. <https://www.education.sa.gov.au/mobile-phones>. Accessed 2025-05-17.
- Department of Education, Tasmania (2019). Use of mobile phones by students at school policy. <https://publicdocumentcentre.education.tas.gov.au/library/Document%20Centre/Use-of-Mobile-Phones-by-Students-at-School-Policy.pdf>. Accessed 2025-05-17.
- Department of Social Services and Melbourne Institute of Applied Economic and Social Research (2024). The Household, Income and Labour Dynamics in Australia (HILDA) Survey, GENERAL RELEASE 23 (Waves 1-23).
- Donati, D., Durante, R., Sobbrío, F., and Zejcirovic, D. (2022). Lost in the net? broadband internet and youth mental health.

- Dowrick, C. and Frances, A. (2013). Medicalising unhappiness: new classification of depression risks more patients being put on drug treatment from which they will not benefit. *bmj*, 347.
- Edwards, B., Barnes, R., Rehill, P., Ellen, L., Zhong, F., Killigrew, A., Gonzalez, P. R., Sheard, E., Zhu, R., and Philips, T. (2022). *Variation in policy response to COVID-19 across Australian states and territories*. Blavatnik School of Government, University of Oxford Oxford, UK.
- Gallup (2023). Teens Spend Average of 4.8 Hours on Social Media Per Day. <https://news.gallup.com/poll/512576/teens-spend-average-hours-social-media-per-day.aspx>. 2024-12-03.
- Gillitzer, C. and Prasad, N. (2024). The effect of school closures on standardized test scores: Evidence under zero-covid policies. *Economics of Education Review*, 102:102577.
- Golin, M. (2022). The effect of broadband Internet on the gender gap in mental health: Evidence from Germany. *Health Economics*, 31:6–21.
- Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2):254–277.
- Government of Western Australia (2019). State-wide mobile phone ban for WA public schools. Media statement.
- Haidt, J. (2024). *The Anxious Generation: How the Great Rewiring of Childhood is Causing an Epidemic of Mental Illness*. Penguin Books, Limited.
- Hale, L. and Guan, S. (2015). Screen time and sleep among school-aged children and adolescents: a systematic literature review. *Sleep medicine reviews*, 21:50–58.
- Hedegaard, H., Curtin, S. C., and Warner, M. (2018). *Suicide rates in the United States continue to increase*, volume 309. US Department of Health and Human Services, Centers for Disease Control and Prevention.
- International Telecommunication Union (2023). Youth Internet Use. <https://www.itu.int/itu-d/reports/statistics/2023/10/10/ff23-youthinternet-use/>. Accessed: May 18, 2025.
- Jensen, M., George, M. J., Russell, M. R., and Odgers, C. L. (2019). Young adolescents’ digital technology use and mental health symptoms: Little evidence of longitudinal or daily linkages. *Clinical Psychological Science*, 7(6):1416–1433.
- Joshi, A. and Hinkley, T. (2021). Too much time on screens? screen time effects and guidelines for children and young people. *Australian Institute of Family Studies*.

- Keles, B., McCrae, N., and Grealish, A. (2020). A systematic review: the influence of social media on depression, anxiety and psychological distress in adolescents. *International Journal of Adolescence and Youth*, 25(1):79–93.
- Kessler, R. C., Barker, P. R., Colpe, L. J., Epstein, J. F., Gfroerer, J. C., Hiripi, E., Howes, M. J., Normand, S.-L. T., Manderscheid, R. W., Walters, E. E., et al. (2003). Screening for serious mental illness in the general population. *Archives of general psychiatry*, 60(2):184–189.
- La Greca, A. M. and Harrison, H. M. (2005). Adolescent peer relations, friendships, and romantic relationships: Do they predict social anxiety and depression? *Journal of Clinical Child and Adolescent Psychology*, 34(1):49–61.
- Lawrence, D., Johnson, S., Hafekost, J., De Haan, K. B., Sawyer, M., Ainley, J., and Zubrick, S. (2015). Report on the second australian child and adolescent survey of mental health and wellbeing. *Department of Health, Australia*.
- Li, S. H., Beames, J. R., Newby, J. M., Maston, K., Christensen, H., and Werner-Seidler, A. (2022). The impact of covid-19 on the lives and mental health of australian adolescents. *European child & adolescent psychiatry*, 31(9):1465–1477.
- McDool, E., Powell, P., Roberts, J., and Taylor, K. (2020). The internet and children’s psychological wellbeing. *Journal of Health Economics*, 69:102274.
- McGorry, P. D., Mei, C., Dalal, N., Alvarez-Jimenez, M., Blakemore, S.-J., Browne, V., Dooley, B., Hickie, I. B., Jones, P. B., McDaid, D., et al. (2024). The lancet psychiatry commission on youth mental health. *The Lancet Psychiatry*, 11(9):731–774.
- McHale, R., Brennan, N., Freeburn, T., and Rossetto, A. (2023). Youth survey report 2023.
- Mojtabai, R., Olfson, M., and Han, B. (2016). National trends in the prevalence and treatment of depression in adolescents and young adults. *Pediatrics*, 138(6).
- Morgan, C., Webb, R. T., Carr, M. J., Kontopantelis, E., Green, J., Chew-Graham, C. A., Kapur, N., and Ashcroft, D. M. (2017). Incidence, clinical management, and mortality risk following self harm among children and adolescents: cohort study in primary care. *bmj*, 359.
- Mosquera, R., Odunowo, M., McNamara, T., Guo, X., and Petrie, R. (2020). The economic effects of Facebook. *Experimental Economics*, 23:575–602.
- Murray, C. J., Vos, T., Lozano, R., Naghavi, M., Flaxman, A. D., Michaud, C., Ezzati, M., Shibuya, K., Salomon, J. A., Abdalla, S., et al. (2012). Disability-adjusted life

- years (dalys) for 291 diseases and injuries in 21 regions, 1990–2010: a systematic analysis for the global burden of disease study 2010. *The Lancet*, 380(9859):2197–2223.
- Northern Territory Department of Education (2022). Student mobile phones in government schools – policy. https://education.nt.gov.au/__data/assets/pdf_file/0003/1171344/student-mobile-phones-in-government-schools.pdf. Accessed 2025-05-17.
- NSW Department of Education (2023). Mobile phones now banned in all nsw public schools. <https://education.nsw.gov.au/news/latest-news/mobile-phones-now-banned-in-all-nsw-public-schools>. Accessed 2025-05-17.
- Odgers, C. L. and Jensen, M. R. (2020). Annual research review: Adolescent mental health in the digital age: Facts, fears, and future directions. *Journal of Child Psychology and Psychiatry*, 61(3):336–348.
- Orben, A. and Przybylski, A. K. (2019). The association between adolescent well-being and digital technology use. *Nature Human Behaviour*, 3(2):173–182.
- Orben, A., Przybylski, A. K., Blakemore, S.-J., and Kievit, R. A. (2022). Windows of developmental sensitivity to social media. *Nature Communications*, 13(1):1649.
- Parliament of the Commonwealth of Australia (2024). Online safety amendment (social media minimum age) bill 2024: Explanatory memorandum. https://parlinfo.aph.gov.au/parlInfo/download/legislation/ems/r7284_ems_b9c134ac-a19a-47b2-9879-b03dda6e3c1a/upload_pdf/JC014726.pdf. Circulated by authority of the Minister for Communications, the Hon Michelle Rowland MP. Accessed: 2025-05-02.
- Parry, D. A., Davidson, B. I., Sewall, C. J., Fisher, J. T., Mieczkowski, H., and Quintana, D. S. (2021). A systematic review and meta-analysis of discrepancies between logged and self-reported digital media use. *Nature Human Behaviour*, 5(11):1535–1547.
- Patton, G. C., Sawyer, S. M., Santelli, J. S., Ross, D. A., Afifi, R., Allen, N. B., Arora, M., Azzopardi, P., Baldwin, W., Bonell, C., et al. (2016). Our future: a lancet commission on adolescent health and wellbeing. *The Lancet*, 387(10036):2423–2478.
- Podsakoff, P. M., MacKenzie, S. B., Lee, J.-Y., and Podsakoff, N. P. (2003). Common method biases in behavioral research: a critical review of the literature and recommended remedies. *Journal of Applied Psychology*, 88(5):879.

- Queensland Government (2023). Mobile Phones in Queensland State Schools Will Be 'Away for the Day'. <https://statements.qld.gov.au/statements/98137>. Accessed 2025-05-17.
- Rhodes, A. (2017). Screen time and kids: What's happening in our homes? Detailed Report Poll 7, The Royal Children's Hospital, Melbourne, Melbourne, Australia. Report amended June 21, 2017.
- Sawyer, S. M., Azzopardi, P. S., Wickremarathne, D., and Patton, G. C. (2018). The Age of Adolescence. *The Lancet Child & Adolescent Health*, 2(3):223–228.
- Selwyn, N. (2019). Digital lessons? public opinions on the use of digital technologies in Australian schools. *Monash University, Education Futures*.
- Solmi, M., Radua, J., Olivola, M., Croce, E., Soardo, L., Salazar de Pablo, G., Il Shin, J., Kirkbride, J. B., Jones, P., Kim, J. H., et al. (2022). Age at onset of mental disorders worldwide: large-scale meta-analysis of 192 epidemiological studies. *Molecular psychiatry*, 27(1):281–295.
- Somerville, L. H. (2013). The teenage brain: Sensitivity to social evaluation. *Current Directions in Psychological Science*, 22(2):121–127.
- Somerville, L. H., Jones, R. M., and Casey, B. (2010). A time of change: behavioral and neural correlates of adolescent sensitivity to appetitive and aversive environmental cues. *Brain and Cognition*, 72(1):124–133.
- State Government of Victoria (2019). Mobile phones to be banned next year in all state schools. Published 26 June 2019. Accessed 20 May 2025.
- Sun, L. and Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of econometrics*, 225(2):175–199.
- Twenge, J. M., Joiner, T. E., Rogers, M. L., and Martin, G. N. (2018). Increases in depressive symptoms, suicide-related outcomes, and suicide rates among US adolescents after 2010 and links to increased new media screen time. *Clinical psychological science*, 6(1):3–17.
- Victorian Department of Education (2019). Mobile phones in schools. <https://www.vic.gov.au/mobile-phones-schools>. Accessed 2025-05-17.
- Vogels, E. A., Gelles-Watnick, R., and Massarat, N. (2022). Teens, social media and technology 2022. <https://www.pewresearch.org/internet/2022/08/10/teens-social-media-and-technology-2022/>. Accessed: 2025-05-23.
- Ware, J. E., Keller, S. D., and Kosinski, M. (1994). *SF-36 Physical and Mental Health Summary Scales*. Health Assessment Lab Boston.

World Economic Forum (2023). Charted: There are more mobile phones than people in the world. <https://www.weforum.org/stories/2023/04/charted-there-are-more-phones-than-people-in-the-world/>. Accessed: 2024-12-03.

World Health Organization (2021). Adolescent mental health. <https://www.who.int/news-room/fact-sheets/detail/adolescent-mental-health>. Accessed: 2024-12-01.

World Health Organization (2025). Suicide. <https://www.who.int/news-room/fact-sheets/detail/suicide>. Accessed: 2025-05-07.

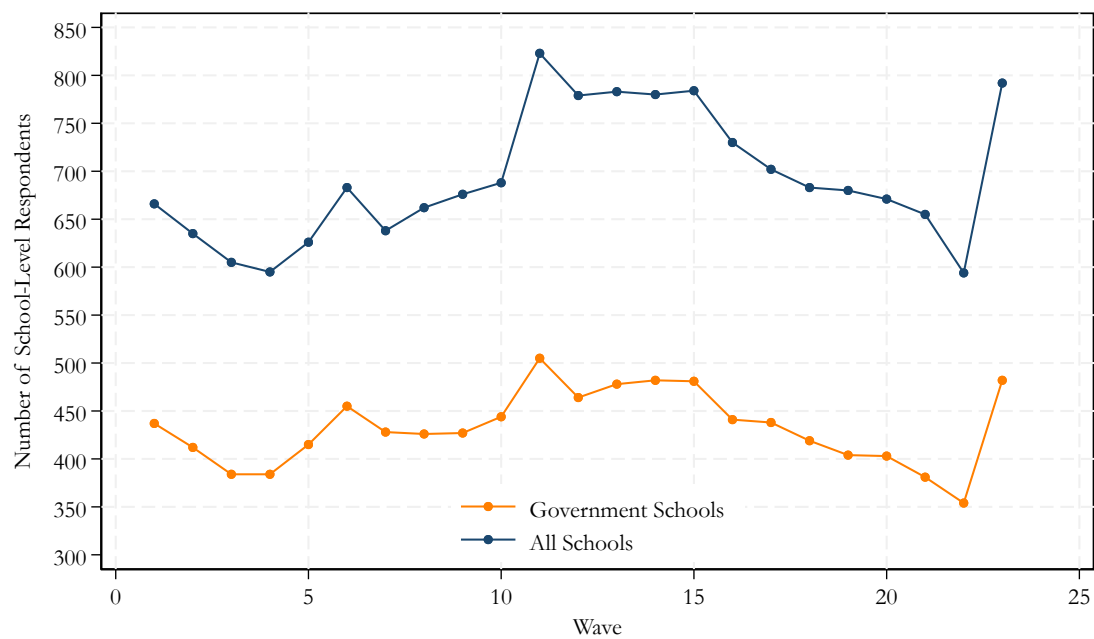
Data Source Acknowledgment

This paper uses unit record data from the Household, Income and Labour Dynamics in Australia (HILDA) Survey. The HILDA Survey was initiated and is funded by the Australian Government Department of Social Services (DSS) and is managed by the Melbourne Institute of Applied Economic and Social Research (Melbourne Institute). The findings and views reported in this paper, however, are those of the author and should not be attributed to the Australian Government, DSS or the Melbourne Institute. DOI: 10.26193/NBTNMV

Appendix

A1 Background and Data

Figure A1. HILDA respondents enrolled in School: Government vs Non-Government Schools Over Time



Notes: This figure shows the number of respondents in each HILDA wave who were enrolled in school at the time of the interview, separately for all school types and for government schools only.

Table A1. SF-36 Mental Health-Related Questions and Response Options

Question	How much of the time during the past 4 weeks...					
	<i>All of the time</i>	<i>Most of the time</i>	<i>A good bit of the time</i>	<i>Some of the time</i>	<i>A little of the time</i>	<i>None of the time</i>
<i>Mental Health</i>						
Have you been a nervous person?	1	2	3	4	5	6
Have you felt so down in the dumps that nothing could cheer you up?	1	2	3	4	5	6
Have you felt calm and peaceful?	1	2	3	4	5	6
Have you felt down?	1	2	3	4	5	6
Have you been a happy person?	1	2	3	4	5	6
<i>Vitality</i>						
Did you feel full of life?	1	2	3	4	5	6
Did you have a lot of energy?	1	2	3	4	5	6
Did you feel worn out?	1	2	3	4	5	6
Did you feel tired?	1	2	3	4	5	6
<i>Social Functioning</i>						
During the past 4 weeks, how much of the time has your physical health or emotional problems interfered with your social activities?						
All / Most / Some / A little / None						
During the past 4 weeks, to what extent has your physical health or emotional problems interfered with your normal social activities?						
Not at all / Slightly / Moderately / Quite a bit / Extremely						
<i>Notes:</i> This table lists the exact questions and response options used to construct the SF-36 Mental Health summary measure.						

Table A2. Start Dates of School Mobile Phone Bans by State and Territory

State / Territory	Ban Start Date	Source
Victoria (VIC)	28 January 2020	VIC Department of Education (2019)
Western Australia (WA)	3 February 2020	Government of Western Australia (2019)
Tasmania (TAS)	27 April 2020	TAS Department of Education (2019)
Northern Territory (NT)	1 January 2023	NT Department of Education (2022)
South Australia (SA)	24 July 2023	SA Department of Education (2023)
New South Wales (NSW)	9 October 2023	NSW Department of Education (2023)
Australian Capital Territory (ACT)	30 January 2024	ACT Government (2023)
Queensland (QLD)	22 January 2024	Queensland Government (2023)

Table A3. Documented School Closure Periods Due to COVID-19 (2020 Expansion Group)

State	Year-level	Region	Year	Term	Closure Start	Closure End
WA	all	Statewide	2020	1	2020-03-30	2020-04-09
WA	< 11	Statewide	2020	2	2020-04-29	2020-05-18
WA	all	Statewide	2021	1	2021-02-01	2021-02-08
TAS	all	Statewide	2020	1	2020-03-25	2020-04-09
TAS	11 & 12	Statewide	2020	2	2020-04-27	2020-05-25
TAS	< 11	Statewide	2020	2	2020-04-27	2020-06-09
TAS	all	Southern Tasmania	2021	4	2021-10-18	2021-10-19
VIC	all	Statewide	2020	1	2020-03-24	2020-03-27
VIC	all	Statewide	2020	2	2020-04-14	2020-06-08
VIC	all	Mitchell Shire & Melbourne	2020	3	2020-07-13	2020-09-18
VIC	all	Rest of VIC	2020	3	2020-08-06	2020-09-18
VIC	all	Statewide	2020	4	2020-10-05	2020-10-09
VIC	all	Statewide	2021	1	2021-02-15	2021-02-17

Notes: School closure dates are collected only for states in the first treatment expansion group. In all remaining states, mobile phone bans were implemented after the final COVID-related school closures had occurred.

Table A4. Summary Statistics for Selected Covariates

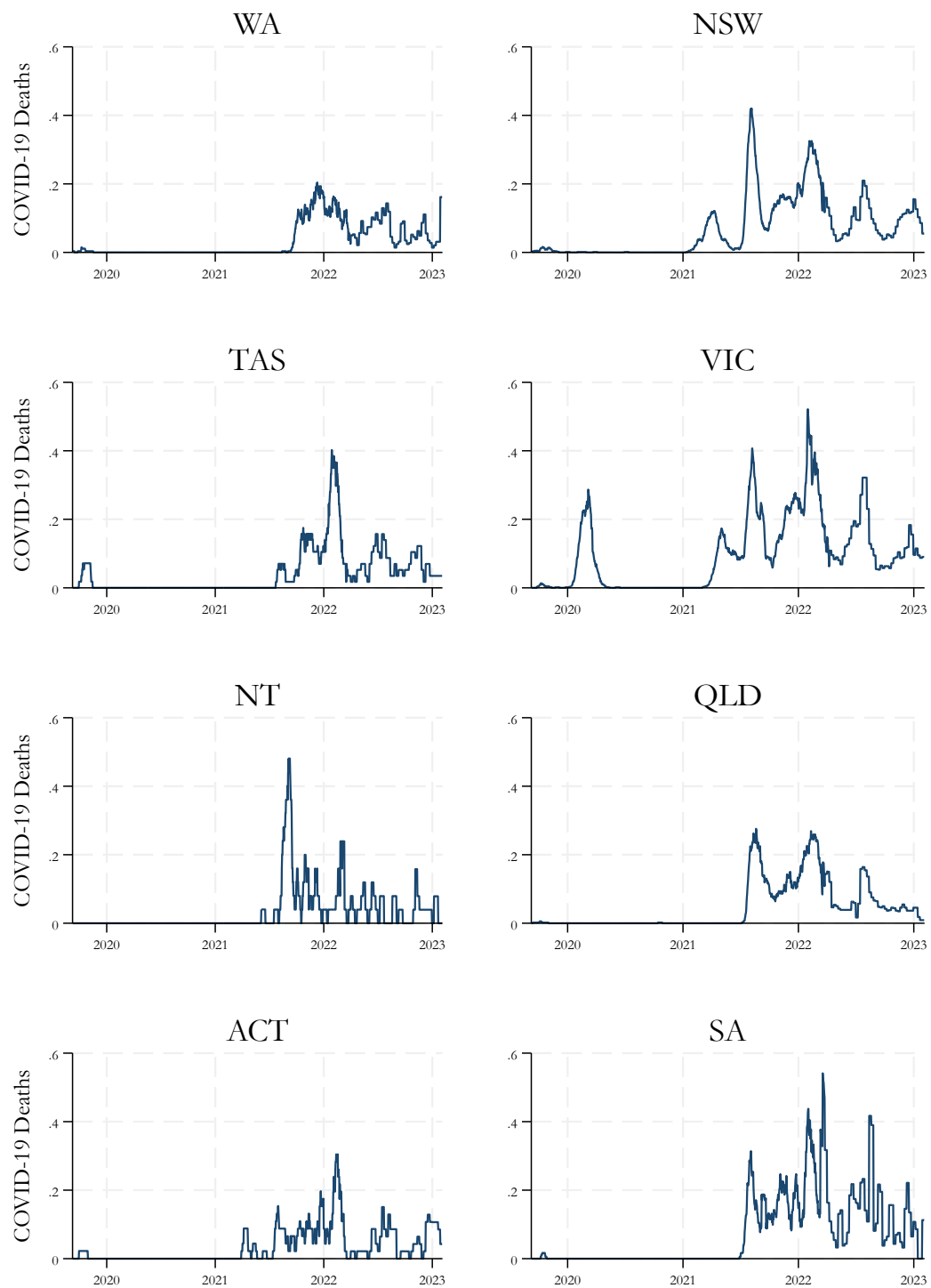
Variable	Mean	SD	N
Individual Characteristics			
Age	16.00	0.95	8320
% Female	50.09	50.01	8320
% Long Term Health Condition	13.98	34.58	8318
Household Characteristics			
Household Income Unit (1-8)	1.06	0.27	8320
Household Education and Occupation decile	4.98	2.76	8320
Household Socio-economic Advantage decile	5.02	2.79	8320
Household Size	4.31	1.45	8320
Household Home Debt (\$)	98,363.35	156,537.43	8090
% Parents Divorced	9.33	28.83	5153
Father weekly gross wages & salaries (\$)	1,035.97	912.30	6101
Mother weekly gross wages & salaries (\$)	545.43	552.50	7528
Unemployment rate in major statistical region	5.30	0.82	8320
SF-36 Health Items			
SF-36 Mental Health Scale (0-100)	72.18	17.60	7547
# been nervous [†]	4.56	1.20	7531
# felt nothing cheers you up [†]	5.16	1.14	7523
# felt calm [†]	2.90	1.20	7531
# felt down [†]	4.65	1.16	7525
# felt happy [†]	2.41	1.09	7513
# physical/emotional health interfered with social activities [‡]	4.39	0.90	7513
Extent physical/emotional health interfered with social activities [§]	1.65	0.94	7515
# felt full of life [†]	2.67	1.17	7513
# felt energetic [†]	2.75	1.22	7513
# felt worn out [†]	4.33	1.22	7499
# felt tired [†]	3.79	1.30	7530
SF-36 General Health Scale (0-100)	73.34	18.79	7526

Notes: Summary statistics are calculated from pre-treatment data. The sample consists in HILDA respondents enrolled in government schools at the time of the interview. Means and standard deviations are first calculated within each survey wave, then averaged across waves. Sample sizes reflect the total number of non-missing observations across all waves. All variables marked with a [†] are rated on a 6-point Likert scale ranging from 1 (= "All the time") to 6 (= "None of the time"). The variable marked with a [‡] is assessed on a 5-point scale ranging from 1 (= "All the time") to 5 (= "None of the time"). The variable marked with a [§] is rated on a 5-point scale ranging from 1 (= "Not at all") to 5 (= "Extremely").

Table A5. Covariate Means by Expansion Group

	Expansion Group		
	2020	2023	Never-Treated
Individual Characteristics			
Age	16.06	16.03	15.84
% Female	50.13	51.15	47.97
% Long Term Health Condition	14.86	13.22	13.95
Household Characteristics			
Household Income Unit (1-8)	1.06	1.05	1.05
Household education and occupation decile	5.22	4.93	4.66
Household socio-economic advantage decile	5.16	5.01	4.80
Household Size	4.31	4.38	4.20
Household home debt (\$)	93,823.71	102,307.09	99,144.79
% Parents Divorced	10.67	8.06	9.90
Father weekly gross wages & salaries (\$)	1,135.09	960.29	1,007.72
Mother weekly gross wages & salaries (\$)	531.07	548.31	572.47
Unemployment rate in major statistical region	5.28	5.37	5.20
SF-36 Health Items			
SF-36 Mental Health Scale (0-100)	71.90	72.44	72.34
# been nervous person [†]	4.53	4.58	4.59
# felt nothing cheers you up [†]	5.15	5.18	5.15
# felt calm [†]	2.90	2.89	2.93
# felt down [†]	4.64	4.65	4.67
# felt happy [†]	2.43	2.41	2.40
# physical/emotional health interfered with social activities [‡]	4.38	4.41	4.40
Extent physical/emotional health interfered with social activities [§]	1.66	1.65	1.64
# felt full of life [†]	2.70	2.64	2.69
# felt energetic [†]	2.77	2.73	2.76
# felt worn out [†]	4.30	4.39	4.28
# felt tired [†]	3.75	3.86	3.70
SF-36 General Health Scale (0-100)	73.11	74.16	72.19

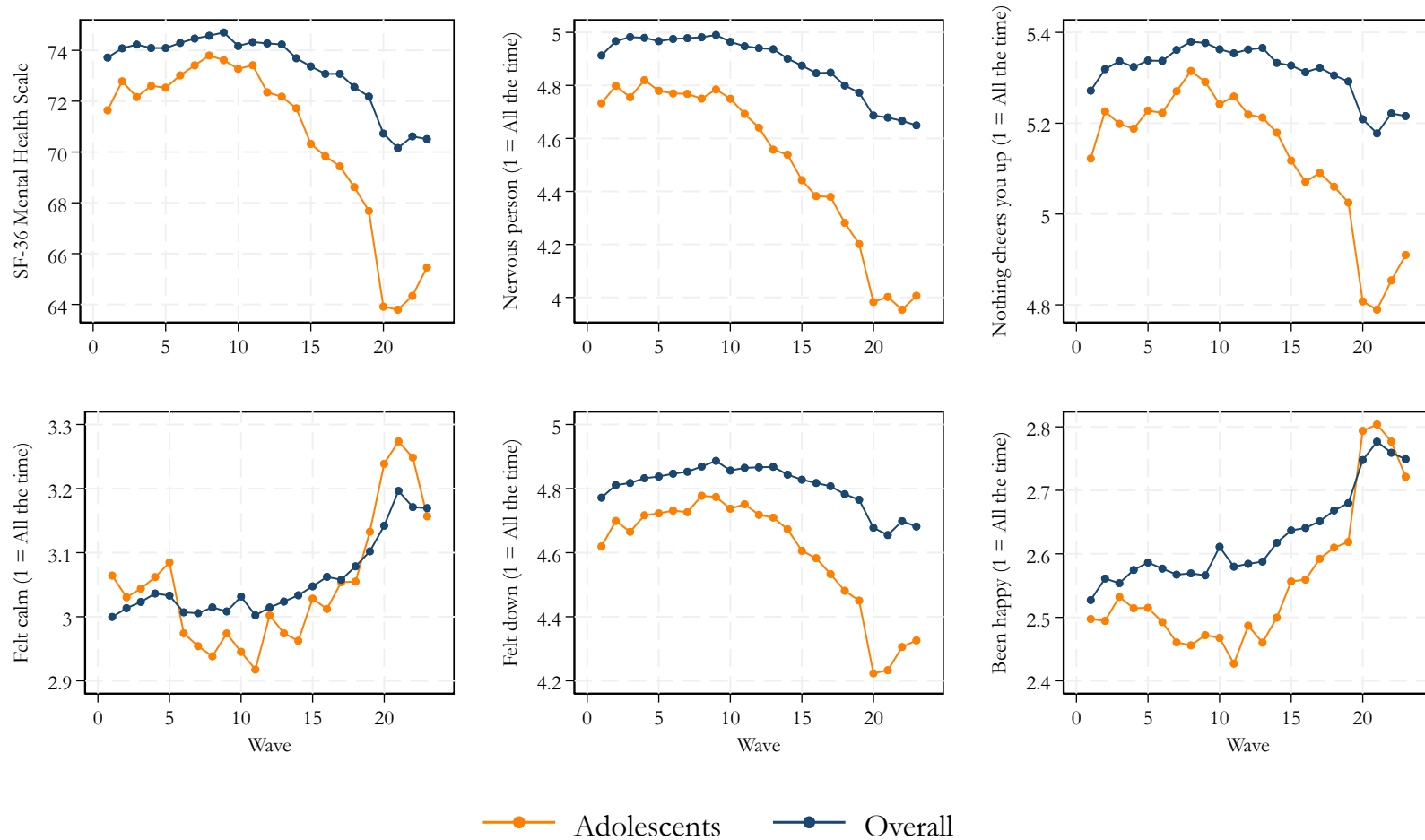
Notes: Summary statistics are calculated from pre-treatment data. The sample consists in HILDA respondents enrolled in government schools at the time of the interview. Means and standard deviations are first calculated within each survey wave, then averaged across waves. Sample sizes reflect the total number of non-missing observations across all waves. All variables marked with a [†] are rated on a 6-point Likert scale ranging from 1 (= "All the time") to 6 (= "None of the time"). The variable marked with a [‡] is assessed on a 5-point scale ranging from 1 (= "All the time") to 5 (= "None of the time"). The variable marked with a [§] is rated on a 5-point scale ranging from 1 (= "Not at all") to 5 (= "Extremely").

Figure A2. Evolution of Daily COVID-19 Mortality by Australian State

Mar 8 2020 – Aug 3 2023

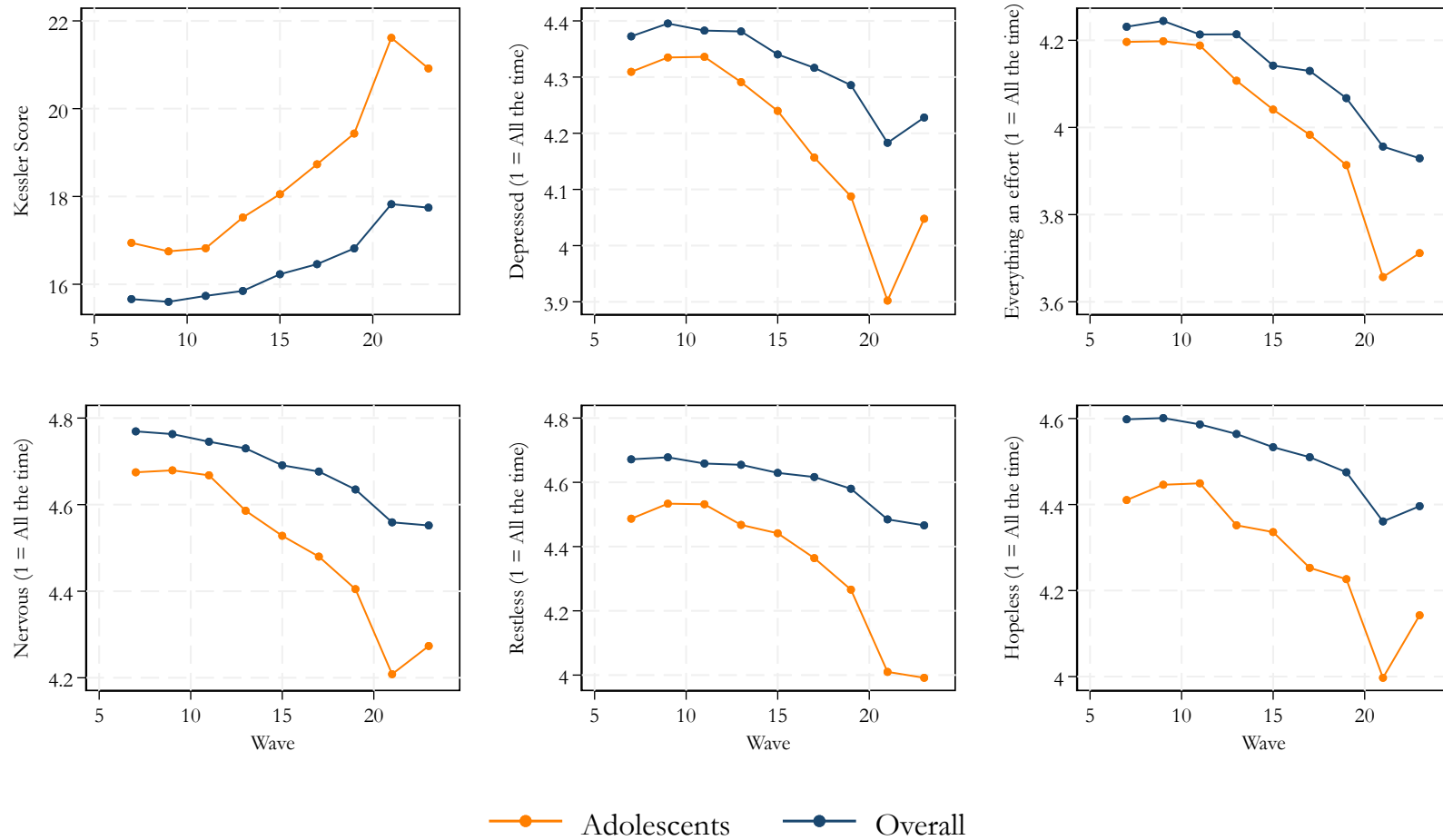
Notes: The figure shows averages of COVID-19 deaths per 100,000 residents across Australian states and territories from March 2020 to August 2023. Death counts are based on a 14-day moving average and scaled using state-specific population data.

Figure A3. Evolution of the SF-36 Mental Health Scale and five underlying Survey Items for Adolescents (15-24)



Notes: This figure shows the evolution of the SF-36 Mental Health score and five of its core components, all of which fall under the *mental health* dimension of the SF-36 questionnaire. Series are plotted separately for all respondents and those enrolled in government schools. Missing values due to non-response or inapplicable items are excluded from the averages.

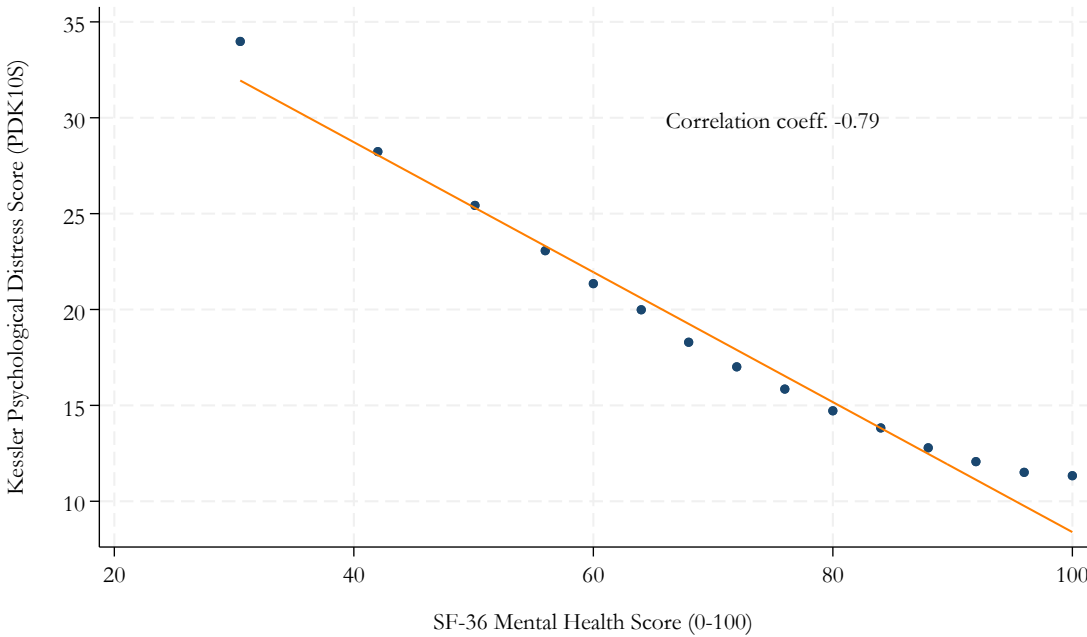
Figure A4. Evolution of all Components of the Kessler Mental Health Score for Adolescents (15-24)



Notes: This figure plots trends in the Kessler Psychological Distress Scale (K10) and its individual components. Separate series are shown for all respondents and adolescents aged 15–24. Averages are calculated excluding missing and inapplicable responses.

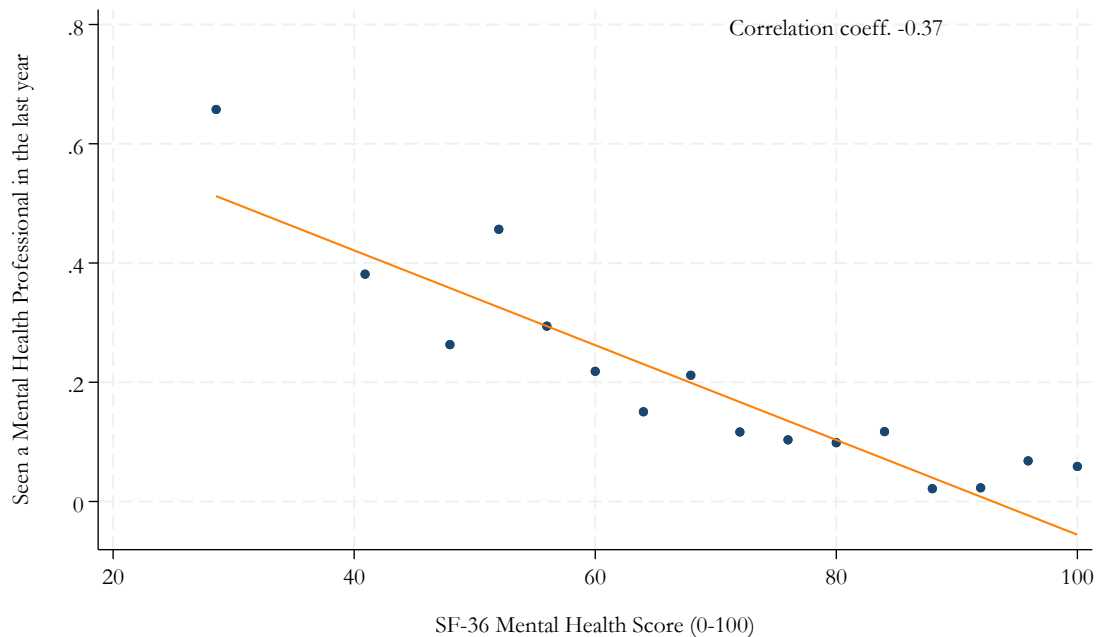
A2 Empirical Strategy

Figure A5. Relationship between SF-36 Mental Health Score and Kessler Scores



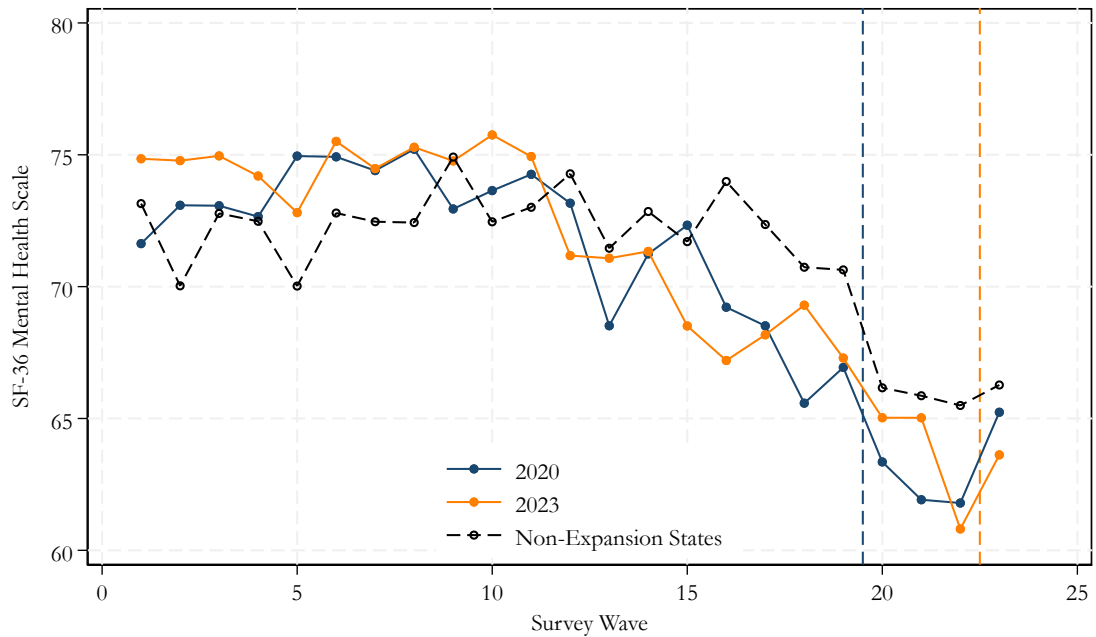
Notes: The figure plots the relationship between the SF-36 Mental Health Score (0–100, higher = better) and the Kessler Psychological Distress Score (10–50, higher = worse) among school-aged respondents. Each dot represents a binned average over 20 quantiles of the SF-36 score. A linear regression line is overlaid. Observations with missing or invalid values in either variable are excluded.

Figure A6. Relationship between SF-36 Mental Health Score and Visits to a Mental Health Professional



Notes: The figure displays the relationship between the SF-36 Mental Health Score (0–100, higher = better) and a binary indicator for having seen a mental health professional in the past year. Each dot represents a binned average over 20 quantiles of the SF-36 score. A linear regression line is overlaid. Observations with missing or invalid values in either variable are excluded.

Figure A7. State Mental Health Trends by Expansion Decision



Notes: The figure plots trends in SF-36 mental health scores by expansion group. Color-coded vertical lines indicate the timing of mobile phone ban implementation for each group. Since bans typically came into effect at the start of the school year, survey responses from the treatment year reflect post-implementation experiences.

A3 Results

Table A6. TWFE Estimates Using Continuous Measure of Treatment Exposure

	SF-36 Mental Health Scale (0-100)		
	(1)	(2)	(3)
<i>Expl. Variables:</i>			
Days of Treatment Exposure	-0.0096 (0.0129)	-0.0011 (0.0099)	-0.0151 (0.0160)
Observations	9,028	9,024	9,027
R ²	0.04109	0.11237	0.75357
Controls		✓	✓
State fixed effects	✓	✓	
Year fixed effects	✓	✓	✓
Individual fixed effects			✓

Notes: All models use a continuous treatment variable capturing the number of days a student was subject to mobile phone bans at the time of interview, accounting for COVID-related school closures and holidays. Standard errors are clustered at the state level. Control variables include student sex, household structure, section of state, long-term health conditions, self-reported pandemic-related disruption, and state-level COVID-19 mortality.

Table A7. Estimated Effects for Male Subsample with Sharpened False Discovery Rate Correction

Variable	Coefficient	<i>p</i> -value	Sharpened <i>q</i> -value
Mental Health			
Nervous	0.1000	0.2332	1.000
Nothing cheers you up	0.0225	0.8215	1.000
Peaceful	0.0527	0.5753	1.000
Down	0.0257	0.6882	1.000
Happy	0.0624	0.4543	1.000
Mental Health Index	0.0710	0.4588	1.000
Vitality			
Full of life	-0.0020	0.9774	1.000
Lot of energy	-0.0594	0.5613	1.000
Worn out	-0.0037	0.9640	1.000
Tired	0.0606	0.4875	1.000
Vitality Index	-0.0006	0.9942	1.000
Social Function			
Emotion & social (i)	0.2451	0.0002	0.004
Emotion & social (ii)	0.0972	0.0199	0.087
Social Function Index	0.1829	0.0009	0.006

Notes: Estimates report the effect of mobile phone bans on standardized SF-36 mental health items and indices, based on a sample of male students. All results derive from TWFE regressions with state and survey-wave fixed effects, and control for individual characteristics and COVID-related disruption. Standard errors are clustered at the state level. Reported *q*-values are adjusted for multiple testing using the sharpened false discovery rate procedure of [Anderson \(2008\)](#).

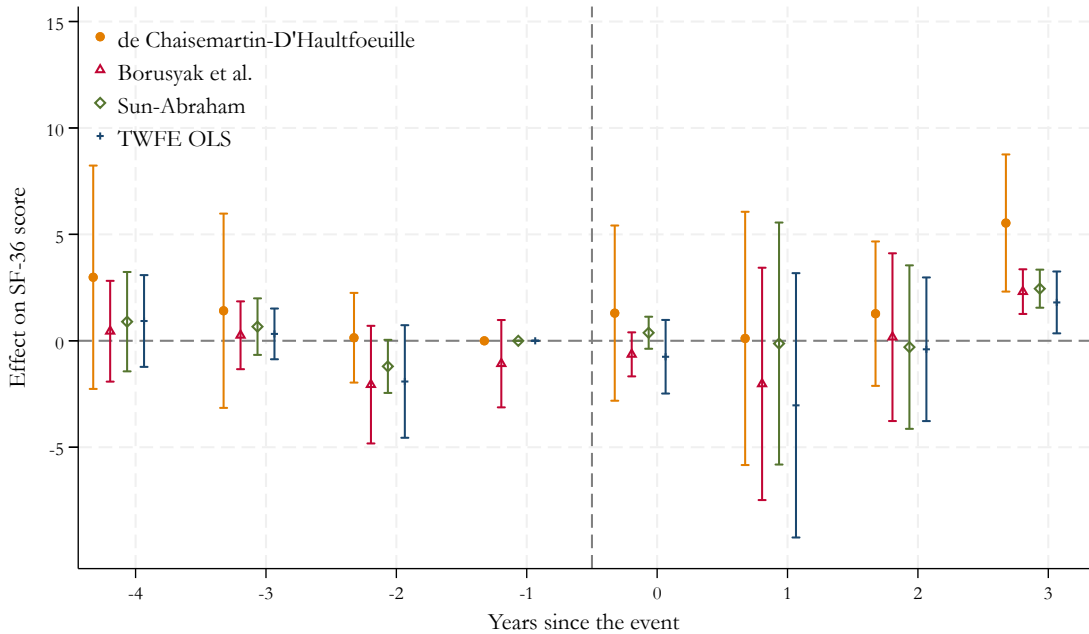
Table A8. Placebo Tests for Pre-Trends

	Estimate	Std. Error	95% CI	N	Switchers
Placebo 1	.1430752	1.047053	[-1.909111, 2.195261]	611	274
Placebo 2	1.41161	2.328854	[-3.152859, 5.976079]	334	131
Placebo 3	2.985857	2.677062	[-2.261088, 8.232802]	325	134

Joint test *p*-value: 0.435

Notes: Placebo tests for the parallel trends assumption as given in [De Chaisemartin and d’Haultfoeuille](#)’s estimation procedure.

Figure A8. Alternative Event Study Estimations: Effects of Mobile Phone Bans on Mental Health



Notes: This figure compares event study estimates using four different event study estimators: TWFE, [Borusyak et al. \(2024\)](#), [Sun and Abraham \(2021\)](#), and [De Chaisemartin and d’Haultfoeuille \(2024\)](#). All models use a binary treatment indicator and include control variables. Error bars represent 95% confidence intervals.

Table A9. Probability of Depression or Anxiety Diagnosis on SF-36 Mental Health Score

<i>Dependent variable</i>	Probability of Depression or Anxiety Diagnosis
	Probit
SF-36 Mental Health Score	-0.0343*** (0.00148)
Constant	2.291*** (0.0972)
Observations	2641
Pseudo R2	0.1764
Likelihood Ratio chi-square(1)	637.99

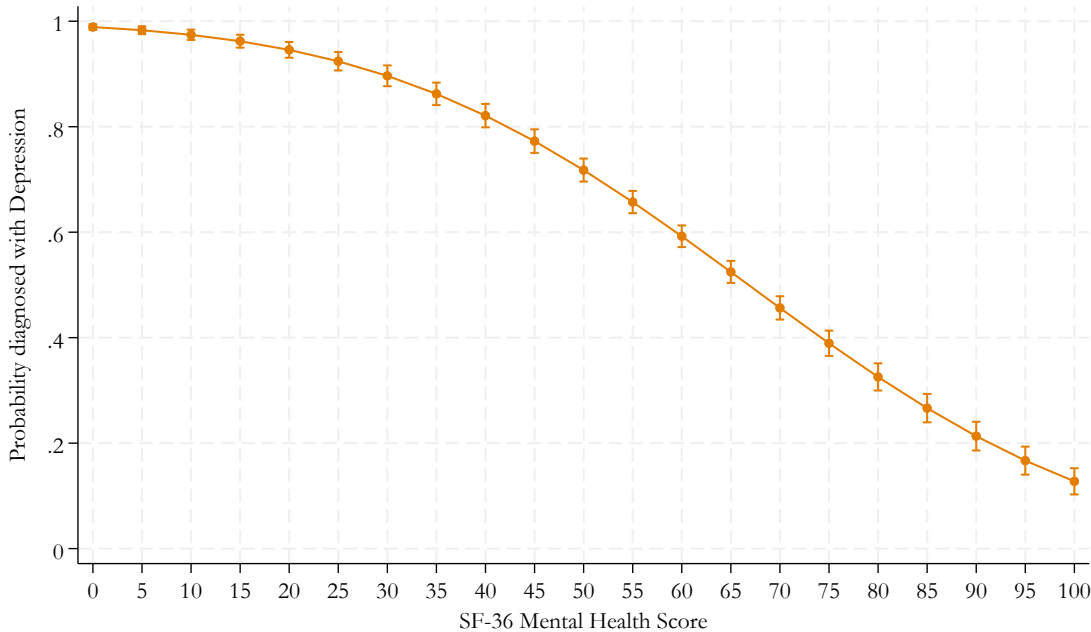
Notes: The table reports coefficients from a probit regression of depression diagnosis (binary outcome) on the SF-36 Mental Health score. The model captures the inverse relationship between better mental health scores and the likelihood of being diagnosed with depression. Standard errors are reported in parentheses.

Table A10. Average Marginal Effect of SF-36 Score on the Probability of Diagnosis

Variable	dy/dx	Std. Error	z-score	p-value	95% CI
SF-36 Mental Health Score	-0.0110	0.0003	-35.25	0.000	[-0.0116, -0.0104]

Notes: This table reports the average marginal effect of the SF-36 Mental Health Score on the probability of being diagnosed with anxiety or depression, based on a probit model. Standard errors are computed using the delta method.

Figure A9. Predicted Probabilities of Clinical Depression or Anxiety Diagnosis along SF-36 scale



Notes: The figure plots average marginal effects of the SF-36 mental health score on the predicted probability of being diagnosed with depression, based on a probit model. The plot illustrates how the marginal impact of an increase in mental health varies across the score distribution. Error bars represent 95% confidence intervals.